SKTC 1623

Thermodynamics

Chapter 7: Entropy

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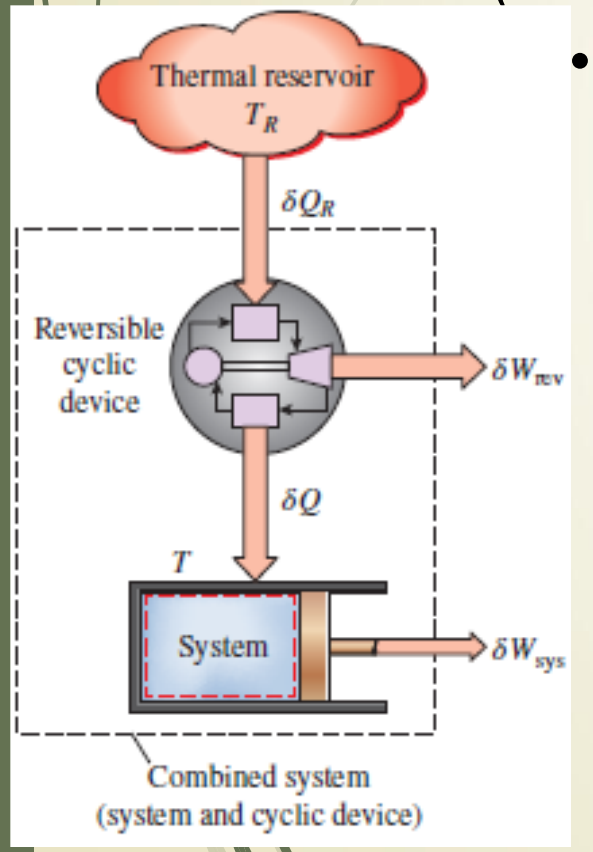
7.1 Entropy and the Clausius Inequality

- The second law of thermodynamics leads to expressions of inequalities.
  - An irreversible heat engine, for example, is less efficient than a reversible one operating between the same two thermal energy reservoirs.
  - Irreversible refrigerator or a heat pump has a lower coefficient of performance (COP) than a reversible one operating between the same temperature limits.



- Another important inequality that has major consequences in thermodynamics is the Clausius inequality, introduced by German physicist R. J. E. Clausius as:

$$\oint \frac{\delta Q}{T} \leq 0$$

the cyclic integral of δQ/T is always less than or equal to zero

  - This inequality is valid for all cycles, reversible or irreversible.
  - The symbol $\oint$ indicate that the integration is to be performed over the entire cycle.
  - Then $\delta Q/T$ can be viewed as the sum of all differential amounts of heat transfer divided by the temperature at the boundary.

7.1 Entropy and the Clausius Inequality (2)

- Consider a heat reservoir giving up heat to a reversible heat engine which in turn gives up heat to a piston-cylinder device as shown in Fig. 7.1
- Apply the first law on an incremental basis to the combined system composed of the heat engine and the system.

$$E_{in} - E_{out} = \Delta E_c$$

$$\delta Q_R - (\delta W_{rev} + \delta W_{sys}) = dE_c$$

where E_c is the energy of the combined system.

- Let W_c be the work done by the combined system.
- Then the first law becomes

$$\delta W_c = \delta W_{rev} + \delta W_{sys}$$

$$\delta Q_R - \delta W_c = dE_c$$

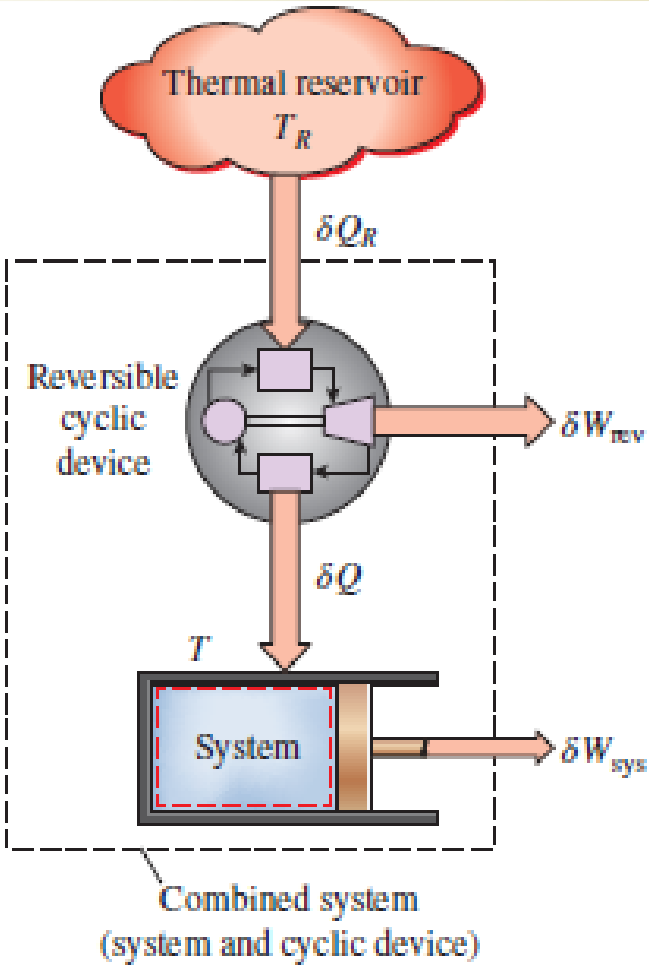


FIGURE 7–1
The system considered in the development of the Clausius inequality.

7.1 Entropy and the Clausius Inequality (3)

- If we assume that the engine is totally reversible, then

$$\frac{\delta Q_R}{T_R} = \frac{\delta Q}{T}$$

For reversible process $\left(\frac{Q_H}{Q_L}\right)_{rev} = \frac{T_H}{T_L}$

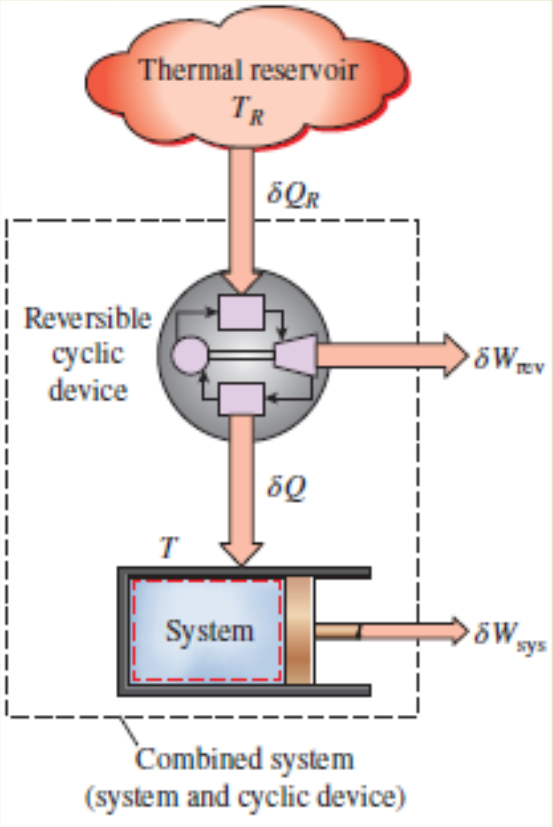
$$\delta Q_R = T_R \frac{\delta Q}{T}$$

- The total net work done by the combined systems becomes

$$\delta W_c = T_R \frac{\delta Q}{T} - dE_c$$

- The total work done is found by taking the cyclic integral of the incremental work.

$$W_c = T_R \oint \frac{\delta Q}{T} - \oint dE_c$$

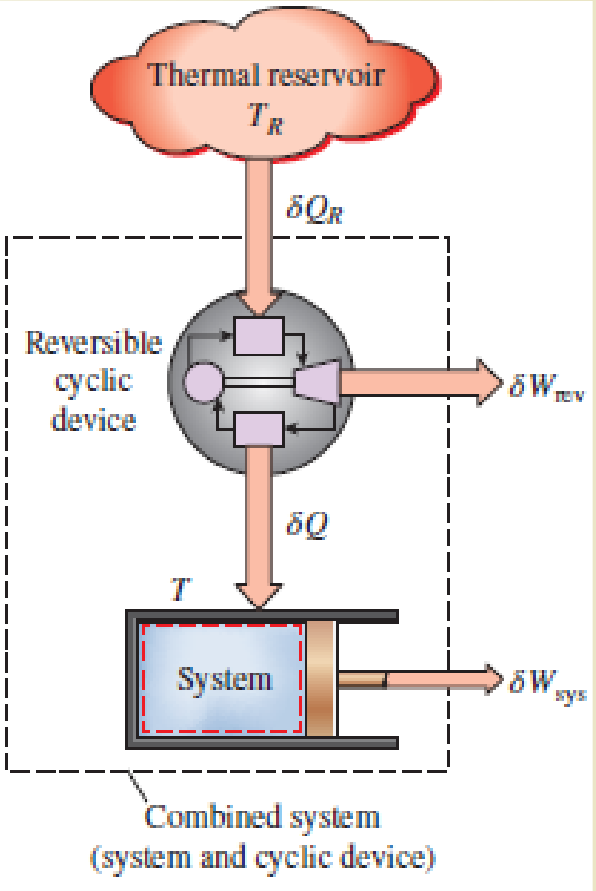


7.1 Entropy and the Clausius Inequality (4)

- Since the cyclic integral of energy (the net change in the energy, which is a property, during a cycle) is zero.

$$\oint dE_c = 0$$

so, the total net work becomes $W_c = T_R \oint \frac{\delta Q}{T}$



- Observing the relation:
  - If W_c is positive,
    - we have a cyclic device exchanging energy with a single heat reservoir and producing an equivalent amount of work;
    - thus, the Kelvin-Planck statement of the second law is violated.
  - $W_c = 0$ indicate there is no work done
  - W_c is negative indicate work is done on the combined system
- The last two do not violate the Kelvin-Planck statement of the second law.

7.1 Entropy and the Clausius Inequality (5)

- Therefore, since $T_R > 0$ (absolute temperature), we conclude

$$\oint \frac{\delta Q}{T} \leq 0$$ *which is the **Clausius Inequality***

- Here the **equality** holds for the **reversible process** and the **inequality** holds for the **irreversible process**.
- Clausius named this new property as **entropy**, designated *S* and defined as

$$dS = \left(\frac{\delta Q}{T}\right)_{\text{int rev}} \qquad \text{(kJ/K)}$$

  - Entropy is an extensive property of a system, sometimes referred as **total entropy**.
  - Entropy per unit mass, *s* is an intensive property and has the unit kJ/kg.K
- **Entropy** can be described as a quantitative measure of microscopic disorder for a system.
  - **Entropy** is a measure of energy that is no longer available to perform useful work within the current environment.

7.1 Entropy and the Clausius Inequality (6)

- The entropy change of a system during a process can be determined by integrating the equation between the initial and final states:

$$\Delta S = S_2 - S_1 = \int_1^2 \left(\frac{\delta Q}{T}\right)_{\text{int rev}} \qquad \text{(kJ/K)}$$

- Note that the integral of $\delta Q/T$ gives the value of entropy change **only if** the integration is carried out along an **internally reversible** path between the two states.
- The integral of $\delta Q/T$ along an irreversible path is not a property, and
  - different values will be obtained when the integration is carried out along different irreversible paths.

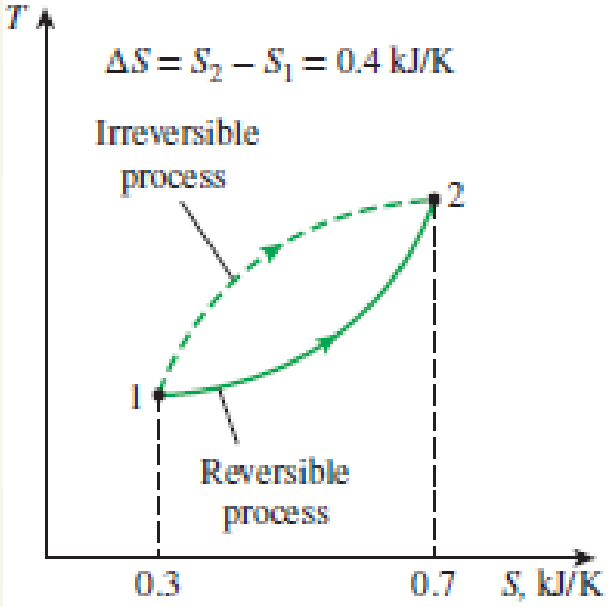


FIGURE 7–3
The entropy change between two specified states is the same whether the process is reversible or irreversible.

7.1 Entropy and the Clausius Inequality (7)

Internally Reversible Isothermal Heat Transfer Processes

- The entropy change of a system during an internally reversible isothermal heat transfer process can be determined by performing the integration:

$$\Delta S = \int_1^2 \left(\frac{\delta Q}{T}\right)_{\text{int rev}} = \int_1^2 \left(\frac{\delta Q}{T_0}\right)_{\text{int rev}} = \frac{1}{T_0}\int_1^2 (\delta Q)_{\text{int rev}}$$

- Which reduces to

$$\Delta S = \frac{Q}{T_0} \qquad (\text{kJ/K})$$

- Where T_0 is the constant temperature of the system and Q is the heat transfer for the internally reversible process.

EXAMPLE 7–1 Entropy Change During an Isothermal Process

A piston–cylinder device contains a liquid–vapor mixture of water at 300 K. During a constant-pressure process, 750 kJ of heat is transferred to the water. As a result, part of the liquid in the cylinder vaporizes. Determine the entropy change of the water during this process.

SOLUTION Heat is transferred to a liquid–vapor mixture of water in a piston–cylinder device at constant pressure. The entropy change of water is to be determined.

Assumptions No irreversibilities occur within the system boundaries during the process.

Analysis We take the *entire water* (liquid + vapor) in the cylinder as the system (Fig. 7–4). This is a *closed system* since no mass crosses the system boundary during the process. We note that the temperature of the system remains constant at 300 K during this process since the temperature of a pure substance remains constant at the saturation value during a phase-change process at constant pressure.

The system undergoes an internally reversible, isothermal process, and thus its entropy change can be determined directly from Eq. 7–6 to be

$$\Delta S_{\text{sys,isothermal}} = \frac{Q}{T_{\text{sys}}} = \frac{750\ \text{kJ}}{300\ \text{K}} = \mathbf{2.5\ kJ/K}$$

Discussion Note that the entropy change of the system is positive, as expected, since heat transfer is to the system.

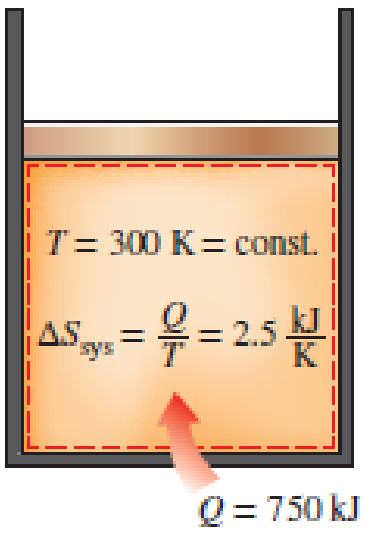


FIGURE 7–4
Schematic for Example 7–1.

7.2 The Increase of Entropy Principle

- Consider a cycle, made up of process 1-2, which is arbitrary (reversible or irreversible), and process 2-1, which is internally reversible, as shown
- From the Clausius inequality,

$$\oint \frac{\delta Q}{T} \leq 0 \quad \text{or} \quad \int_1^2 \frac{\delta Q}{T} + \int_2^1 \left(\frac{\delta Q}{T}\right)_{\text{int rev}} \leq 0$$

- The second integral is the entropy change $S_1 - S_2$.

$$\int_1^2 \frac{\delta Q}{T} + S_1 - S_2 \leq 0$$

which can be rearranged as $$S_2 - S_1 \geq \int_1^2 \frac{\delta Q}{T}$$

- It can also be expressed in differential form as

$$dS \geq \frac{\delta Q}{T}$$

where **equality** holds for an **internally reversible process** and the **inequality** for an **irreversible process**.

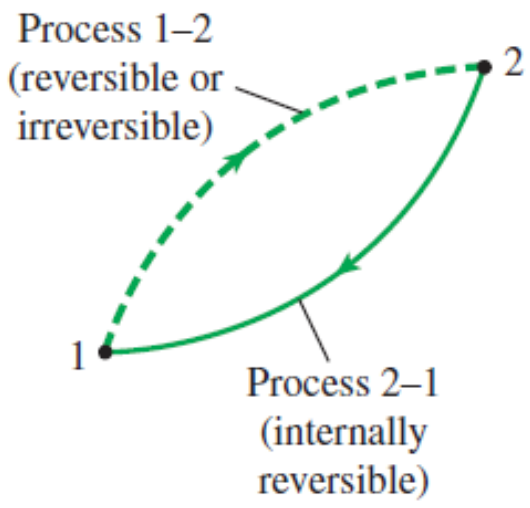


FIGURE 7–5
A cycle composed of a reversible and an irreversible process.

7.2 The Increase of Entropy Principle (2)

$$S_2 - S_1 \geq \int_1^2 \frac{\delta Q}{T}$$

- The quantity $\Delta S = S_2 - S_1$ represents the entropy change of the system.
  - For a reversible process, it becomes equal to $\int_1^2 \delta Q/T$, which represents the entropy transfer with heat.
  - For an irreversible process, it is greater than the integral $\delta Q/T$ evaluated for that process.
- T in this relation is the thermodynamic temperature at the boundary where the differential heat δQ is transferred between the system and the surroundings.
- The entropy change of a closed system during an irreversible process is always greater than the entropy transfer.
- That is, some entropy is generated or created during an irreversible process, and this generation is due entirely to the presence of irreversibilities.

7.2 The Increase of Entropy Principle (3)

- The entropy generated during a process is called **entropy generation** and is denoted as S_{gen}.
- Note that the difference between the entropy change of a closed system and the entropy transfer is equal to entropy generation, that is

$$\Delta S_{sys} = S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} + S_{gen}$$

- The entropy generation S_{gen} is always a positive quantity or zero.
  - Its value depends on the process, and thus it is not a property of the system.
- For an **isolated system** (or simply an adiabatic closed system), the heat transfer is zero, and the entropy change equation reduces to

$$\Delta S_{isolated} \geq 0$$

- This is known as the **increase of entropy principle** which states that

The entropy of an isolated system during a process always increases or, in the limiting case of a reversible process, remains constant.

7.2 The Increase of Entropy Principle (4)

- Note that in the absence of any heat transfer, entropy change is due to irreversibilities only, and their effect is always to increase entropy.
  - In other words, it never decreases.
- Consider an isolated system consists of a system and its surroundings.
  - The entropy change of this isolated system during a process is the sum of the entropy changes of the system and its surroundings, which is equal to the **entropy generation** since an isolated system involves no entropy transfer.

$$S_{gen} = \Delta S_{total} = \Delta S_{sys} + \Delta S_{surr} \geq 0$$

    where the **equality** holds for **reversible processes** and the **inequality** for **irreversible**.
- ΔS_{surr} refers to the change in the entropy of the surroundings as a result of the occurrence of the process.

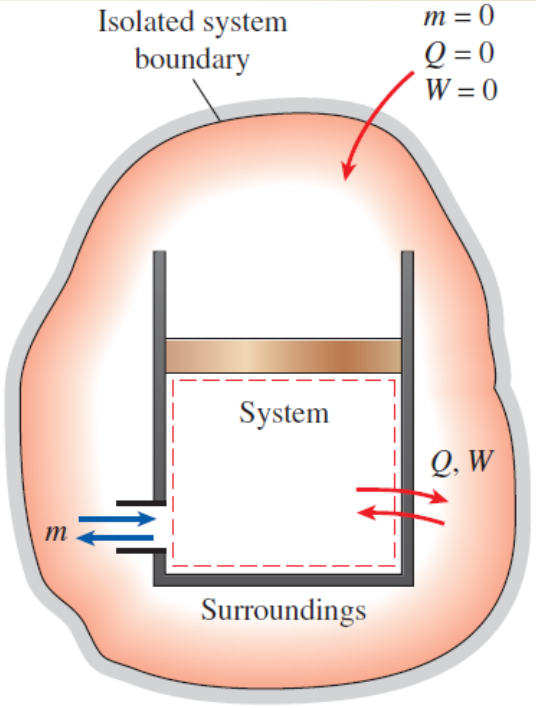
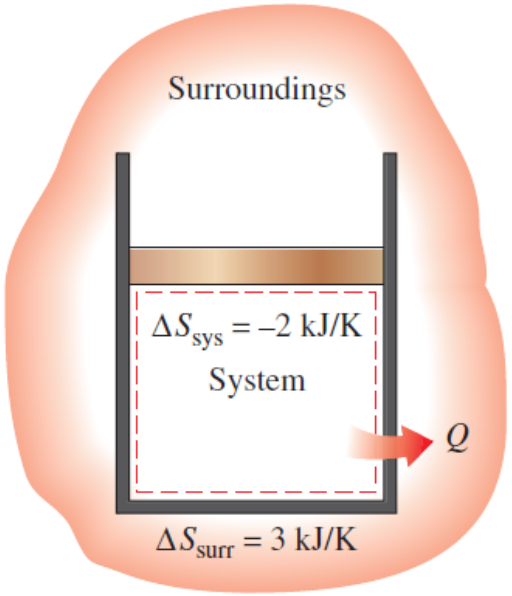


FIGURE 7–7
A system and its surroundings form an isolated system.

7.2 The Increase of Entropy Principle (5)

- Since no actual process is truly reversible, we can conclude that
  - Some entropy is generated during a process, and therefore the entropy of the universe is continuously increasing.
  - The more irreversible a process, the larger the entropy generated during that process.



$S_{gen} = \Delta S_{total} = \Delta S_{sys} + \Delta S_{surr} = 1$ kJ/K

FIGURE 7–8
The entropy change of a system can be negative, but the entropy generation cannot.

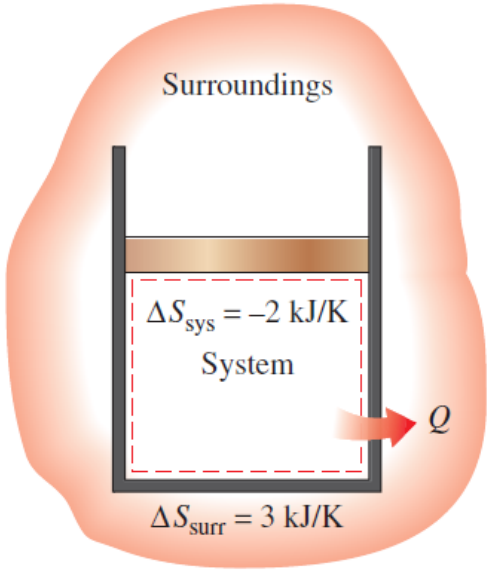
- No entropy is generated during reversible processes ($S_{gen} = 0$).
- Entropy change is caused by heat transfer and irreversibilities.
- Heat transfer to a system increases the entropy, and heat transfer from a system decreases it.
- The effects of irreversibilities is always to increase the entropy.
- The increase in entropy principle can be summarized as follows:

$$S_{gen} \begin{cases} > 0 & \text{irreversible process} \\ = 0 & \text{reversible process} \\ < 0 & \text{impossible process} \end{cases}$$

7.2 The Increase of Entropy Principle (6)

Some Remarks about Entropy

- Processes can occur in a certain direction only, not in any direction such that $S_{gen} \geq 0$.
- Entropy is a **non-conserved property**, and there is no such thing as the conservation of entropy principle.
  - Entropy is conserved during the idealized reversible processes only and increases during all actual processes.
  - The entropy of the universe is continuously increasing.
- The performance of engineering systems is degraded by the presence of irreversibilities.
  - The greater the extent of irreversibilities, the greater the entropy generation.
  - Entropy generation can be used as a quantitative measure of irreversibilities associated with a process.



$S_{gen} = \Delta S_{total} = \Delta S_{sys} + \Delta S_{surr} = 1$ kJ/K

FIGURE 7–8

The entropy change of a system can be negative, but the entropy generation cannot.

EXAMPLE 7–2 Entropy Generation During Heat Transfer Processes

A heat source at 800 K loses 2000 kJ of heat to a sink at (*a*) 500 K and (*b*) 750 K. Determine which heat transfer process is more irreversible.

SOLUTION Heat is transferred from a heat source to two heat sinks at different temperatures. The heat transfer process that is more irreversible is to be determined.

Analysis A sketch of the reservoirs is shown in Fig. 7–9. Both cases involve heat transfer through a finite temperature difference, and therefore both are irreversible. The magnitude of the irreversibility associated with each process can be determined by calculating the total entropy change for each case. The total entropy change for a heat transfer process involving two reservoirs (a source and a sink) is the sum of the entropy changes of each reservoir since the two reservoirs form an adiabatic system.

Or do they? The problem statement gives the impression that the two reservoirs are in direct contact during the heat transfer process. But this cannot be the case since the temperature at a point can have only one value, and thus it cannot be 800 K on one side of the point of contact and 500 K on the other side. In other words, the temperature function cannot have a jump discontinuity. Therefore, it is reasonable to assume that the two reservoirs are separated by a partition through which the temperature drops from 800 K on one side to 500 K (or 750 K) on the other. In that case, the entropy change of the partition should also be considered when evaluating the total entropy change for this process. However, considering that entropy is a property and the values of properties depend on the state of a system, we can argue that the entropy change of the partition is zero since the partition appears to have undergone a *steady* process and thus experienced no change in its properties at any point. We base this argument on the fact that the temperature on both sides of the partition and thus throughout remains constant during this process. Therefore, we are justified to assume that $\Delta S_{partition} = 0$ since the entropy (as well as the energy) content of the partition remains constant during this process.

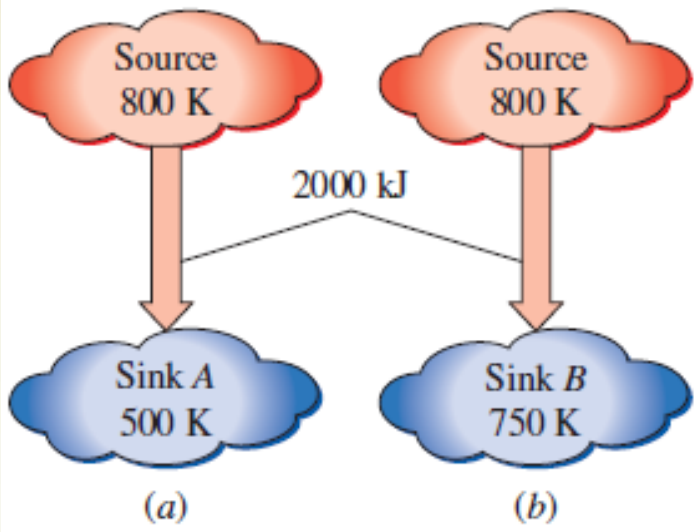


FIGURE 7–9
Schematic for Example 7–2.

The entropy change for each reservoir can be determined from Eq. 7–6 since each reservoir undergoes an internally reversible, isothermal process.

(*a*) For the heat transfer process to a sink at 500 K:

$$\Delta S_{source} = \frac{Q_{source}}{T_{source}} = \frac{-2000\ \text{kJ}}{800\ \text{K}} = -2.5\ \text{kJ/K}$$

$$\Delta S_{sink} = \frac{Q_{sink}}{T_{sink}} = \frac{2000\ \text{kJ}}{500\ \text{K}} = 4.0\ \text{kJ/K}$$

and

$$S_{gen} = \Delta S_{total} = \Delta S_{source} + \Delta S_{sink} = (-2.5 + 4.0)\ \text{kJ/K} = \mathbf{1.5\ kJ/K}$$

Therefore, 1.5 kJ/K of entropy is generated during this process. Noting that both reservoirs have undergone internally reversible processes, the entire entropy generation took place in the partition.

(*b*) Repeating the calculations in part (*a*) for a sink temperature of 750 K, we obtain

$$\Delta S_{source} = -2.5\ \text{kJ/K}$$
$$\Delta S_{sink} = 2.7\ \text{kJ/K}$$

and

$$S_{gen} = \Delta S_{total} = (-2.5 + 2.7)\ \text{kJ/K} = \mathbf{0.2\ kJ/K}$$

The total entropy change for the process in part (*b*) is smaller, and therefore it is less irreversible. This is expected since the process in (*b*) involves a smaller temperature difference and thus a smaller irreversibility.

Discussion The irreversibilities associated with both processes could be eliminated by operating a Carnot heat engine between the source and the sink. For this case it can be shown that $\Delta S_{total} = 0$.

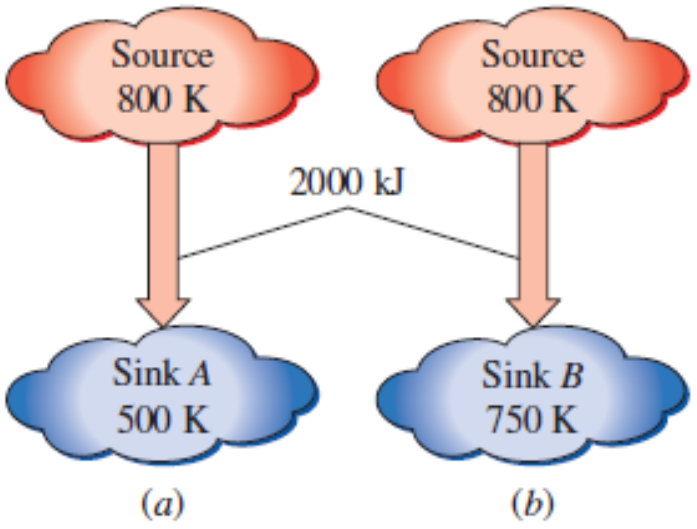


FIGURE 7–9
Schematic for Example 7–2.

7.3 Entropy Change of Pure Substances

- Entropy is a property, and thus the value of entropy of a system is fixed once the state of the system is fixed.
- The entropy values in the property tables are given relative to an arbitrary reference state, similar to other properties in the table.
- In the compressed liquid and superheated vapor regions directly from the tables at the specified state.
- In the saturated mixture region, it is determined from

$$s = s_f + x s_{fg} \qquad \text{(kJ/kg·K)}$$

- The entropy change of a specified mass *m* (a closed system) during a process is

$$\Delta S = m\Delta s = m(s_2 - s_1) \qquad \text{(kJ/K)}$$

- When studying the second-law aspects of processes, entropy is also commonly used as a coordinate on diagrams such as the ***T-s*** and ***h-s*** **diagrams**.

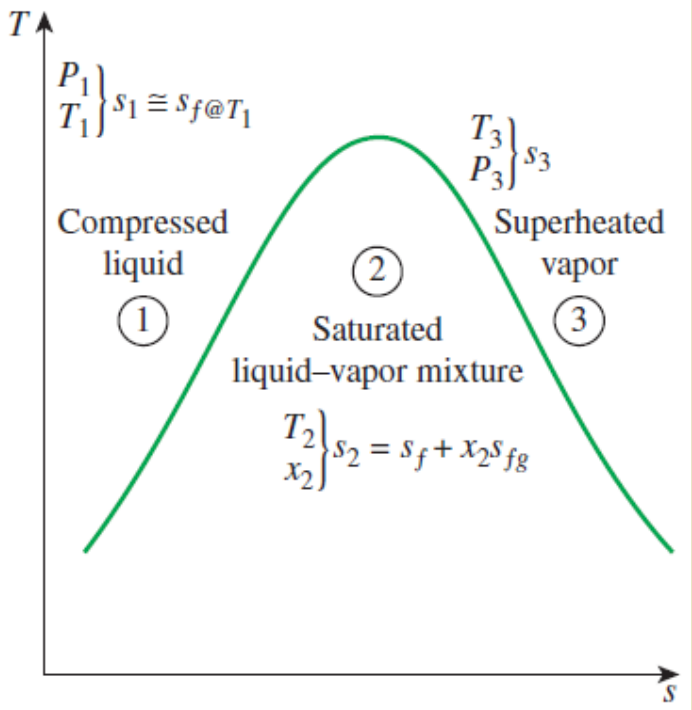


FIGURE 7–10
The entropy of a pure substance is determined from the tables (like other properties).

7.3 Entropy Change of Pure Substances (2)

T-s diagram

- The general characteristics of the *T-s* diagram of pure substances are as shown
- Some observations:
  - *v*-const. lines are steeper than the *p*-const. lines
  - *p*-const. lines are parallel to *T*-const. lines in liquid-vapor mixture region.
  - *p*-const. lines almost coincide saturated liquid lines in compressed region.

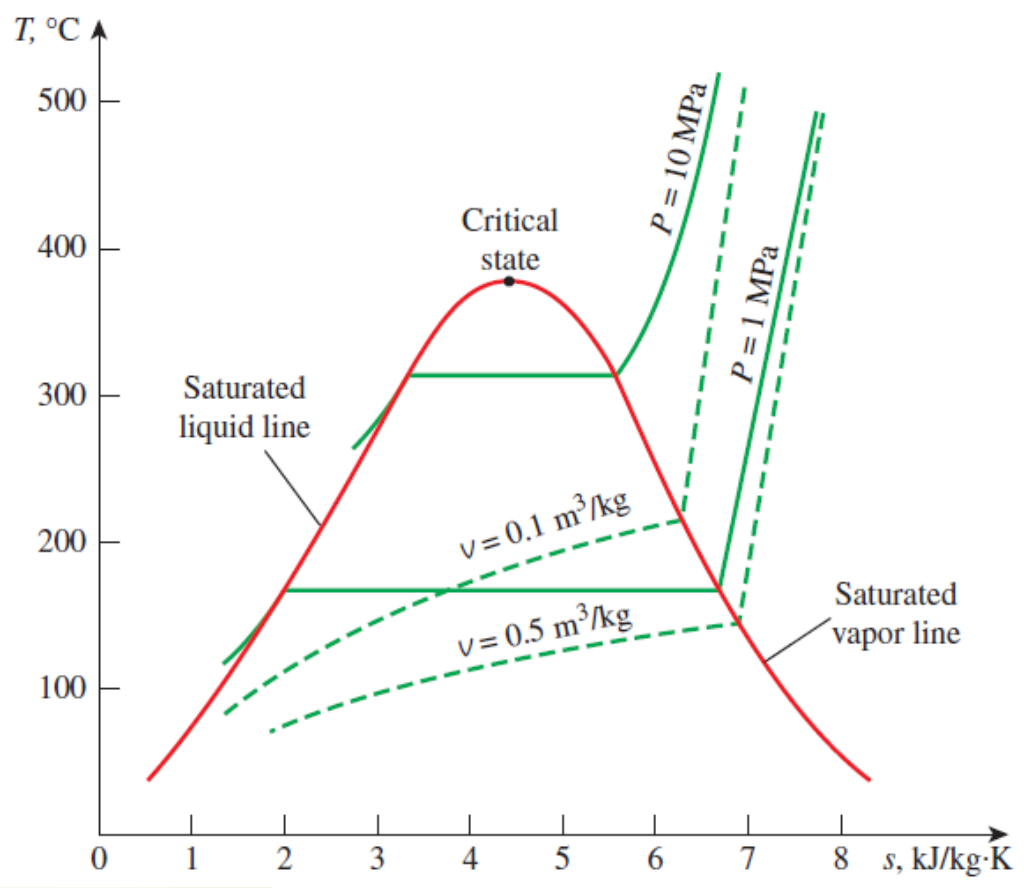


FIGURE 7–11
Schematic of the *T-s* diagram for water.

EXAMPLE 7–3 Entropy Change of a Substance in a Tank

A rigid tank contains 5 kg of refrigerant-134a initially at 20°C and 140 kPa. The refrigerant is now cooled while being stirred until its pressure drops to 100 kPa. Determine the entropy change of the refrigerant during this process.

SOLUTION The refrigerant in a rigid tank is cooled while being stirred. The entropy change of the refrigerant is to be determined.

Assumptions The volume of the tank is constant and thus $v_2 = v_1$.

Analysis We take the refrigerant in the tank as the system (Fig. 7–12). This is a *closed system* since no mass crosses the system boundary during the process. We note that the change in entropy of a substance during a process is simply the difference between the entropy values at the final and initial states. The initial state of the refrigerant is completely specified.

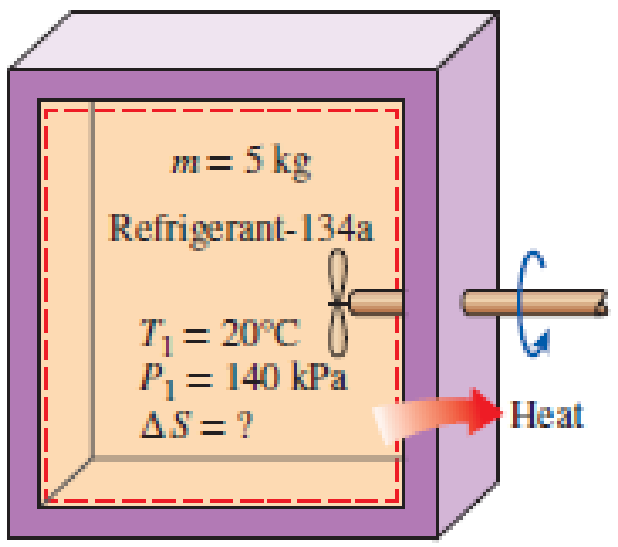
Recognizing that the specific volume remains constant during this process, the properties of the refrigerant at both states are

State 1:

$$\left.\begin{aligned} P_1 &= 140 \text{ kPa} \\ T_1 &= 20°\text{C} \end{aligned}\right\} \begin{aligned} s_1 &= 1.0625 \text{ kJ/kg·K} \\ v_1 &= 0.16544 \text{ m}^3\text{/kg} \end{aligned}$$

State 2:

$$\left.\begin{aligned} P_2 &= 100 \text{ kPa} \\ v_2 &= v_1 \end{aligned}\right\} \begin{aligned} v_f &= 0.0007258 \text{ m}^3\text{/kg} \\ v_g &= 0.19255 \text{ m}^3\text{/kg} \end{aligned}$$



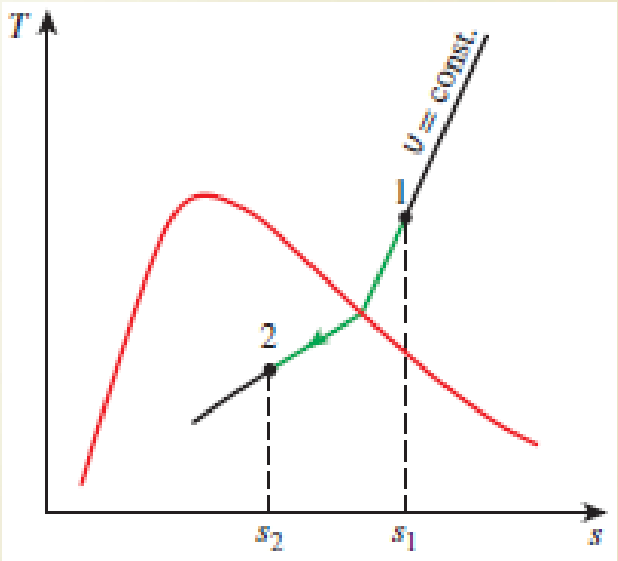


FIGURE 7–12
Schematic and *T-s* diagram for Example 7–3.

The refrigerant is a saturated liquid–vapor mixture at the final state since $v_f < v_2 < v_g$ at 100 kPa pressure. Therefore, we need to determine the quality first:

$$x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.16544 - 0.0007258}{0.19255 - 0.0007258} = 0.859$$

Thus,

$$s_2 = s_f + x_2 s_{fg} = 0.07182 + (0.859)(0.88008) = 0.8278 \text{ kJ/kg·K}$$

Then, the entropy change of the refrigerant during this process is

$$\Delta S = m(s_2 - s_1) = (5 \text{ kg})(0.8278 - 1.0625) \text{ kJ/kg·K}$$
$$= \mathbf{-1.173 \text{ kJ/K}}$$

Discussion The negative sign indicates that the entropy of the system is decreasing during this process. This is not a violation of the second law, however, since it is the *entropy generation* S_{gen} that cannot be negative.

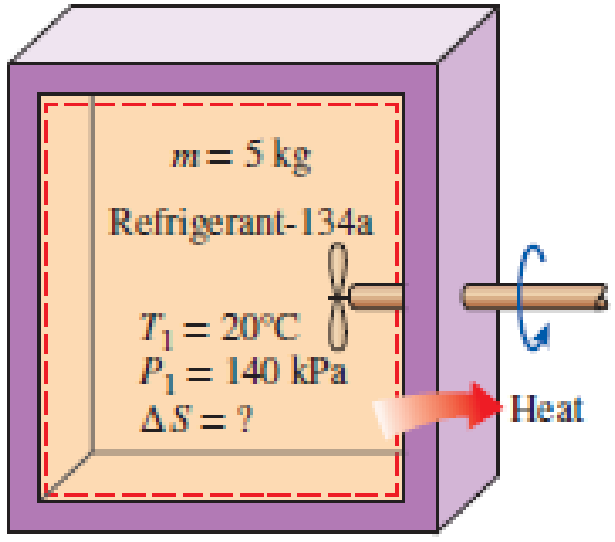
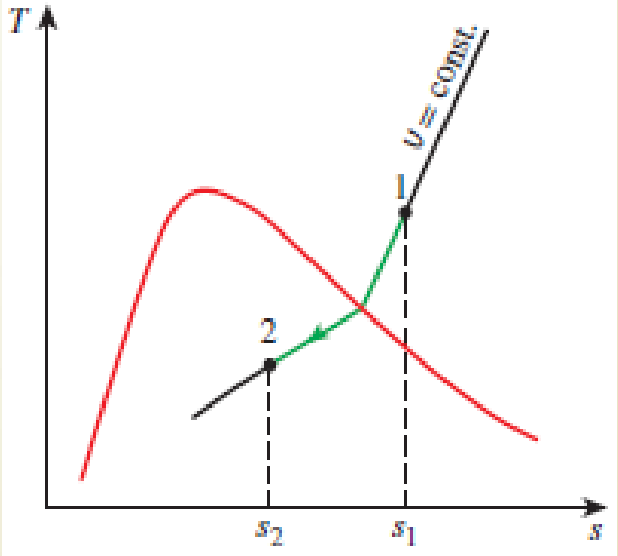


FIGURE 7–12
Schematic and *T-s* diagram for Example 7–3.



7.4 Isentropic Processes

- The entropy of a fixed mass can be changed by
  - heat transfer, and
  - irreversibilities.
- So, the entropy of a fixed mass does not change during a process that is internally reversible and adiabatic.
- A process during which the entropy remains constant is called an isentropic process.
- It is characterized by

$$\Delta s = 0 \quad \text{or} \quad s_2 = s_1 \qquad (\text{kJ/kg·K})$$ *Isentropic process*

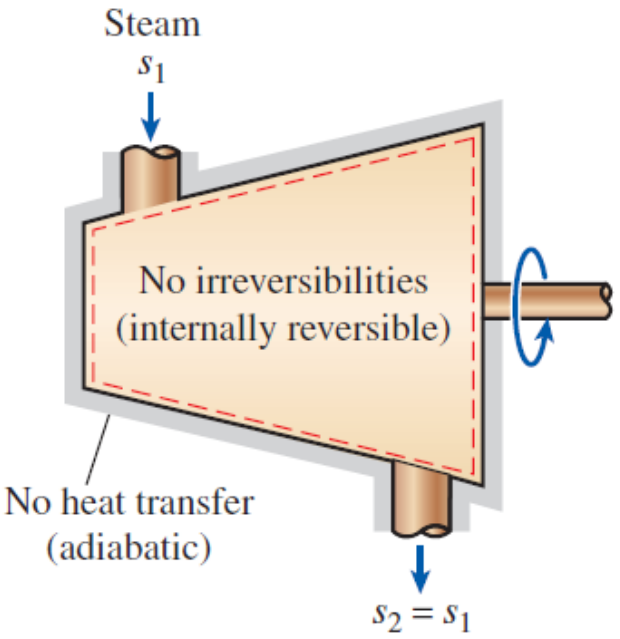


FIGURE 7–14
During an internally reversible, adiabatic (isentropic) process, the entropy remains constant.

- Many engineering systems or devices such as pumps, turbines, nozzles, and diffusers are essentially adiabatic in their operation.
  - Thus, an isentropic process can serve as an appropriate model for actual processes.
  - Isentropic processes enable us to define efficiencies for processes to compare the actual performance of these devices to the performance under idealized conditions.

EXAMPLE 7–5 Isentropic Expansion of Steam in a Turbine

Steam enters an adiabatic turbine at 5 MPa and 450°C and leaves at a pressure of 1.4 MPa. Determine the work output of the turbine per unit mass of steam if the process is reversible.

SOLUTION Steam is expanded in an adiabatic turbine to a specified pressure in a reversible manner. The work output of the turbine is to be determined.

Assumptions **1** This is a steady-flow process since there is no change with time at any point and thus $\Delta m_{CV} = 0$, $\Delta E_{CV} = 0$, and $\Delta S_{CV} = 0$. **2** The process is reversible. **3** Kinetic and potential energies are negligible. **4** The turbine is adiabatic and thus there is no heat transfer.

Analysis We take the *turbine* as the system (Fig. 7–15). This is a *control volume* since mass crosses the system boundary during the process. We note that there is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$.

The power output of the turbine is determined from the rate form of the energy balance,

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\substack{\text{Rate of net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{dE_{system}/dt}_{\substack{\text{Rate of change in internal, kinetic,}\\ \text{potential, etc., energies}}} \nearrow 0\ (\text{steady}) = 0$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}h_1 = \dot{W}_{out} + \dot{m}h_2 \quad (\text{since } \dot{Q} = 0, \text{ ke} \cong \text{pe} \cong 0)$$

$$\dot{W}_{out} = \dot{m}(h_1 - h_2)$$

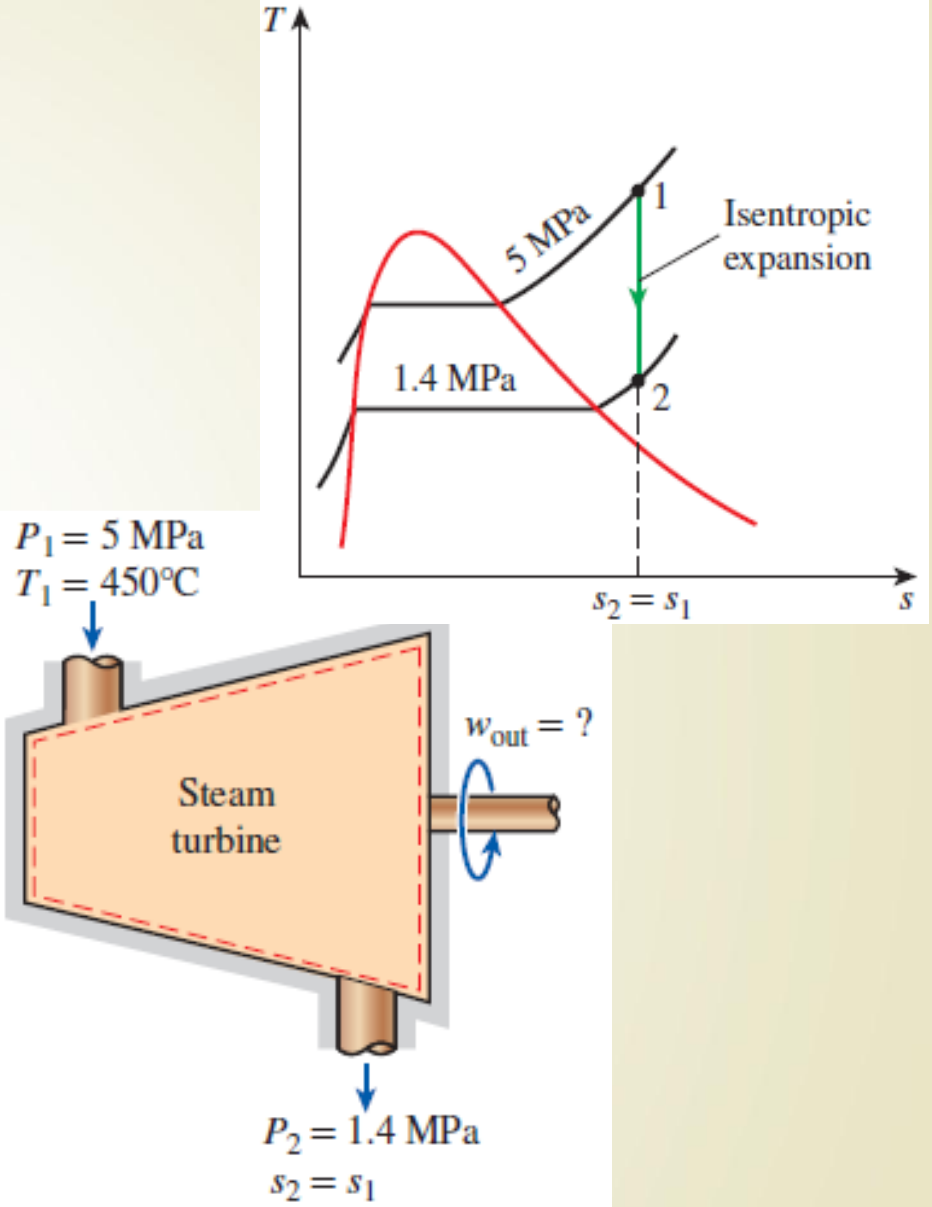


FIGURE 7–15
Schematic and *T-s* diagram for Example 7–5.

The inlet state is completely specified since two properties are given. But only one property (pressure) is given at the final state, and we need one more property to fix it. The second property comes from the observation that the process is reversible and adiabatic, and thus isentropic. Therefore, $s_2 = s_1$, and

State 1:
$$\left.\begin{aligned} P_1 &= 5\ \text{MPa} \\ T_1 &= 450°\text{C} \end{aligned}\right\} \begin{aligned} h_1 &= 3317.2\ \text{kJ/kg} \\ s_1 &= 6.8210\ \text{kJ/kg·K} \end{aligned}$$

State 2:
$$\left.\begin{aligned} P_2 &= 1.4\ \text{MPa} \\ s_2 &= s_1 \end{aligned}\right\} h_2 = 2967.4\ \text{kJ/kg}$$

Then, the work output of the turbine per unit mass of the steam becomes

$$w_{out} = h_1 - h_2 = 3317.2 - 2967.4 = \mathbf{349.8\ kJ/kg}$$

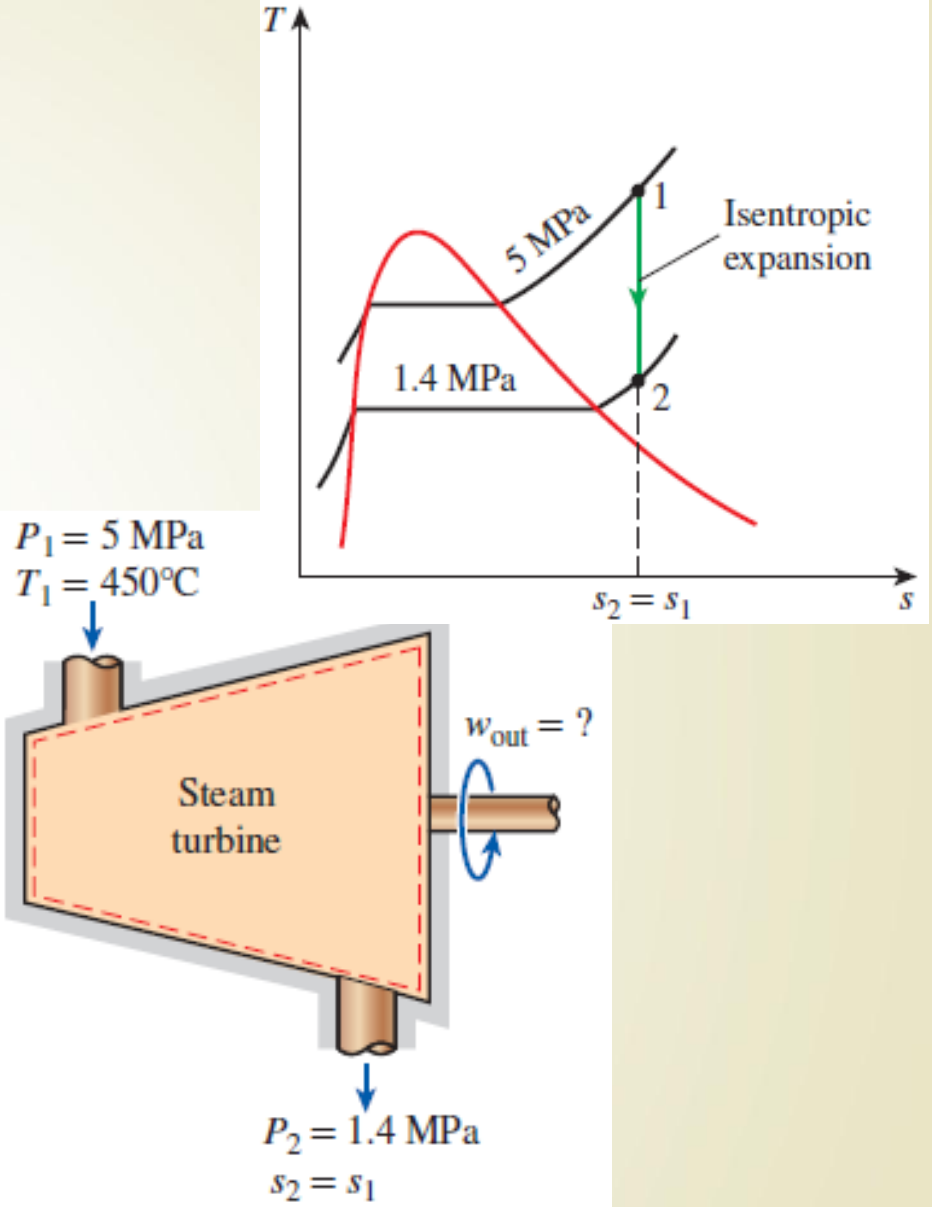


FIGURE 7–15
Schematic and *T-s* diagram for Example 7–5.

7.5 Property Diagrams Involving Entropy

- Property diagrams serve as great visual aids in the thermodynamic analysis of processes.
- *P-v* and *T-v* diagrams are extensively with the first law of thermodynamics.
- In the second-law analysis, it is very helpful to plot the processes on diagrams for which one of the coordinates is entropy.
- The two diagrams commonly used in the second law are
  - the ***temperature-entropy*** (*T-S*) diagram, and
  - the ***enthalpy-entropy*** (*h-s*) diagram (also called Mollier diagram) .
- Recall the previous equation of entropy

$$dS = \left(\frac{\delta Q}{T}\right)_{\text{int rev}} \qquad \text{(kJ/K)}$$

- It can be rearranged as

$$\delta Q_{\text{int rev}} = T\,dS \qquad \text{(kJ)}$$

7.5 Property Diagrams Involving Entropy (2)

T-S diagram

- As shown in figure, $\delta Q_{\text{int rev}}$ corresponds to a differential area on a *T-S* diagram.
- The total heat transfer during an internally reversible process is determined by integration:

$$Q_{\text{int rev}} = \int_1^2 T\,dS \qquad \text{(kJ)}$$

- The area under the curve represents heat transfer for processes during an **internally reversible process**.
  - The area has no meaning for irreversible processes.
- The relations can also be expressed on a unit-mass basis as $\delta q_{\text{int rev}} = T\,ds \qquad \text{(kJ/kg)}$

and $$q_{\text{int rev}} = \int_1^2 T\,ds \qquad \text{(kJ/kg)}$$

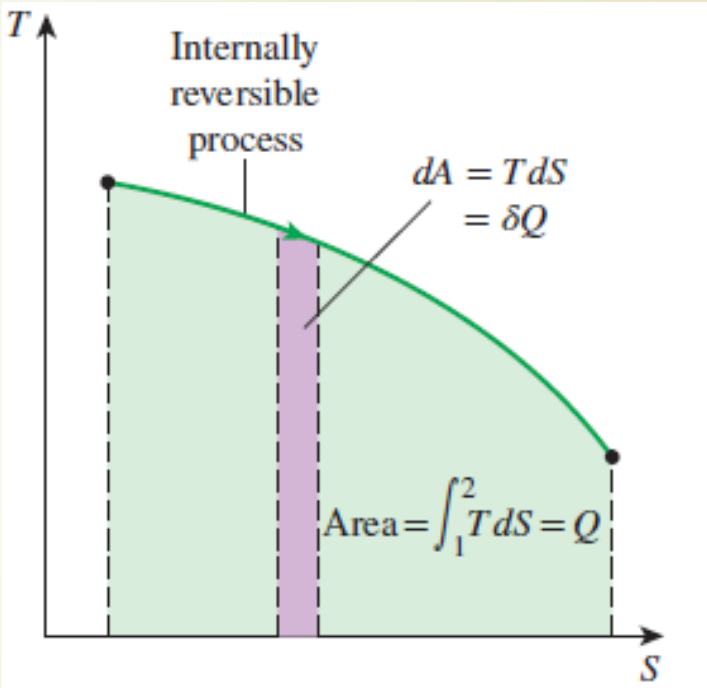


FIGURE 7–16
On a *T-S* diagram, the area under the process curve represents the heat transfer for internally reversible processes.

7.5 Property Diagrams Involving Entropy (3)

- For internally reversible isothermal process

$$Q_{\text{int rev}} = T_0 \, \Delta S \qquad \text{(kJ)}$$

or

$$q_{\text{int rev}} = T_0 \, \Delta s \qquad \text{(kJ/kg)}$$

where T_0 is the constant temperature and ΔS is the entropy change of the system during the process.

- An isentropic process on a *T-s* diagram is easily recognized as a vertical-line segment.
  - This is expected since an isentropic process involves no heat transfer.
  - Therefore, the area under the process path must be zero.
- The *T-s* diagrams serve as valuable tools for visualizing the second-law aspects of processes and cycles, and thus they are frequently used in thermodynamics.

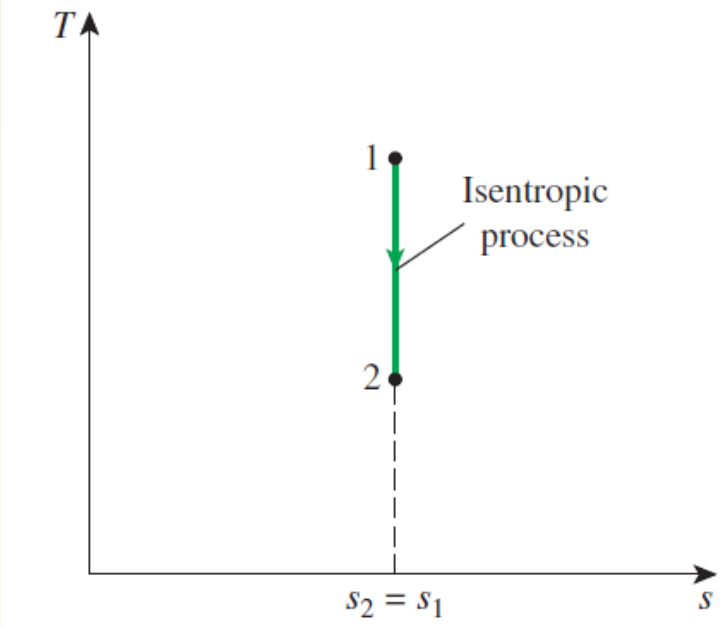


FIGURE 7–17
The isentropic process appears as a vertical line segment on a *T-s* diagram.

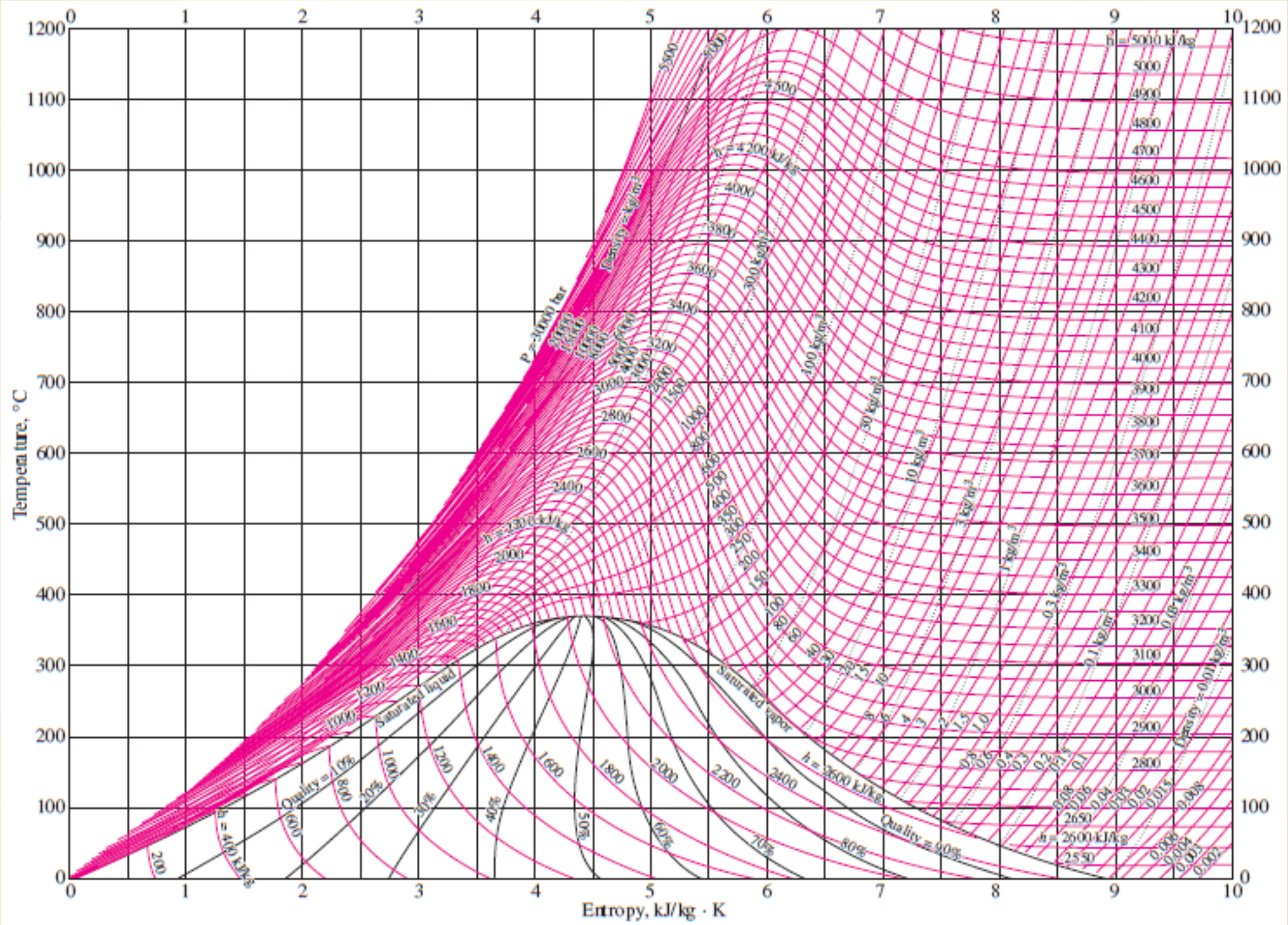
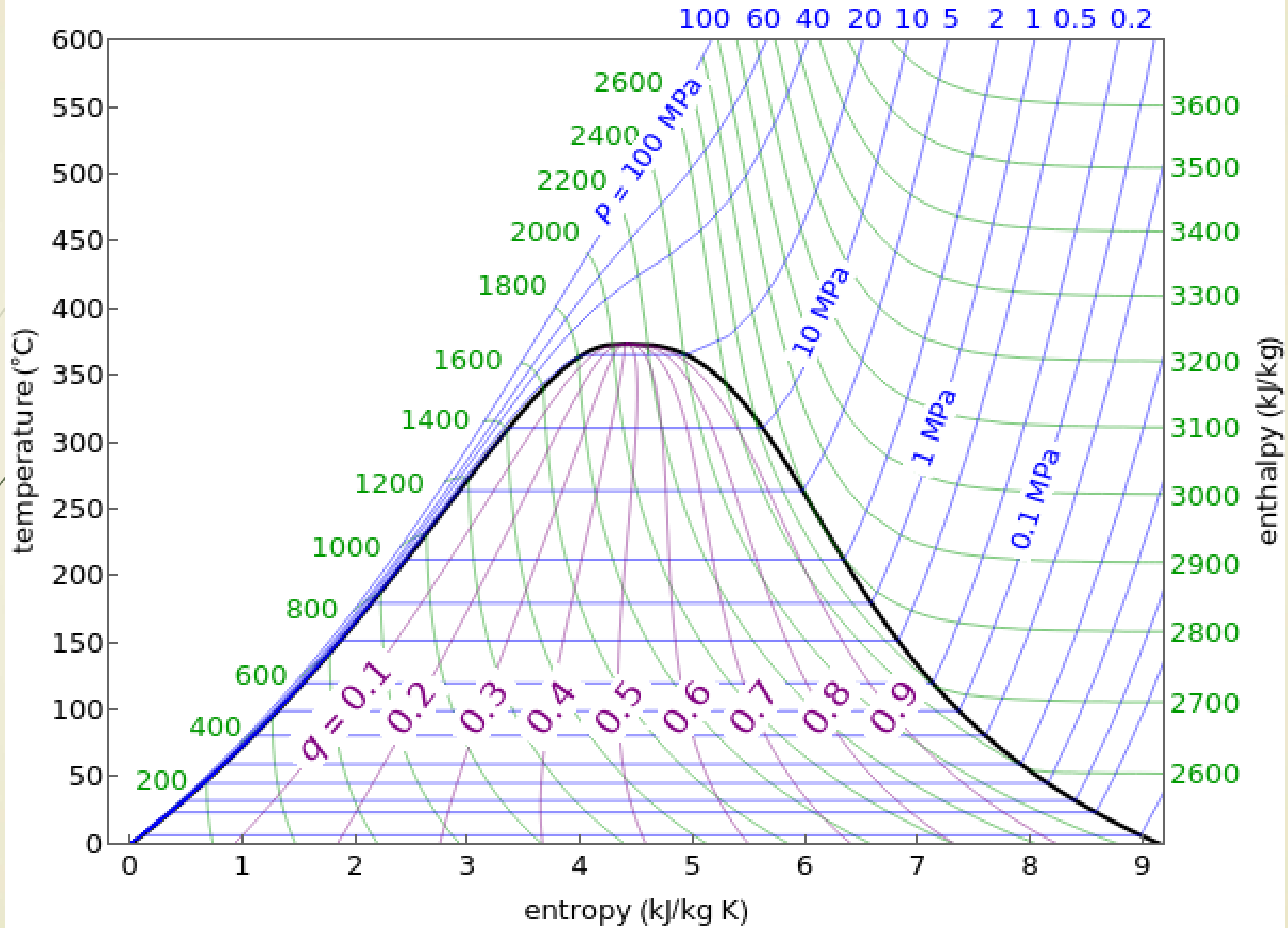


FIGURE A–9

T-s diagram for water.


100 60 40 20 10 5 2 1 0.5 0.2
p = 100 MPa
10 MPa
1 MPa
0.1 MPa
2600
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
q = 0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
temperature (°C)
600
550
500
450
400
350
300
250
200
150
100
50
0
entropy (kJ/kg K)
0 1 2 3 4 5 6 7 8 9
enthalpy (kJ/kg)
3600
3500
3400
3300
3200
3100
3000
2900
2800
2700
2600

7.5 Property Diagrams Involving Entropy (4)

h-s diagram (Mollier diagram)

- Another diagram commonly used is the ***enthalpy-entropy*** (*h-s*) diagram.
- It is quite valuable in the analysis of steady-flow devices such as turbines, compressors, and nozzles.
- The coordinates of an *h-s* diagram represent two properties of major interest:
  - Enthalpy – a primary property in the first-law analysis of the steady-flow devices,
  - Entropy – the property that accounts for irreversibilities during adiabatic processes.
- In analyzing the steady flow of steam through an adiabatic turbine, for example,
  - Δh is a measure of the work output of the turbine,
  - Δs is a measure of the irreversibilities associated with the process

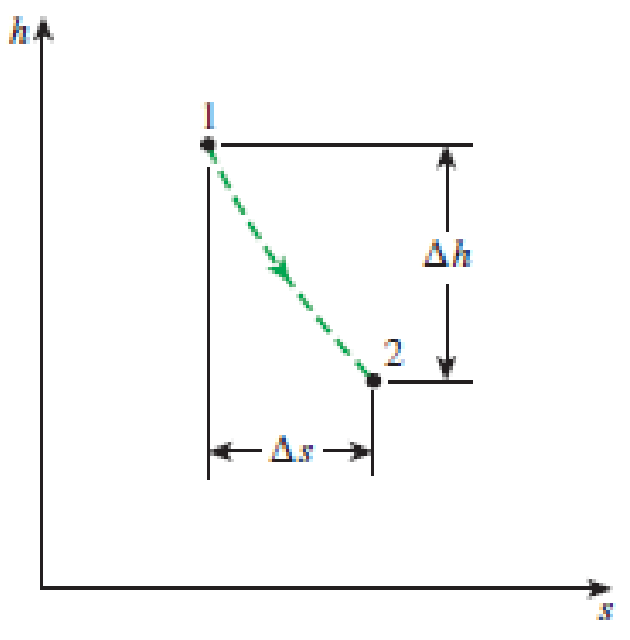


FIGURE 7–18
For adiabatic steady-flow devices, the vertical distance Δh on an *h-s* diagram is a measure of work, and the horizontal distance Δs is a measure of irreversibilities.

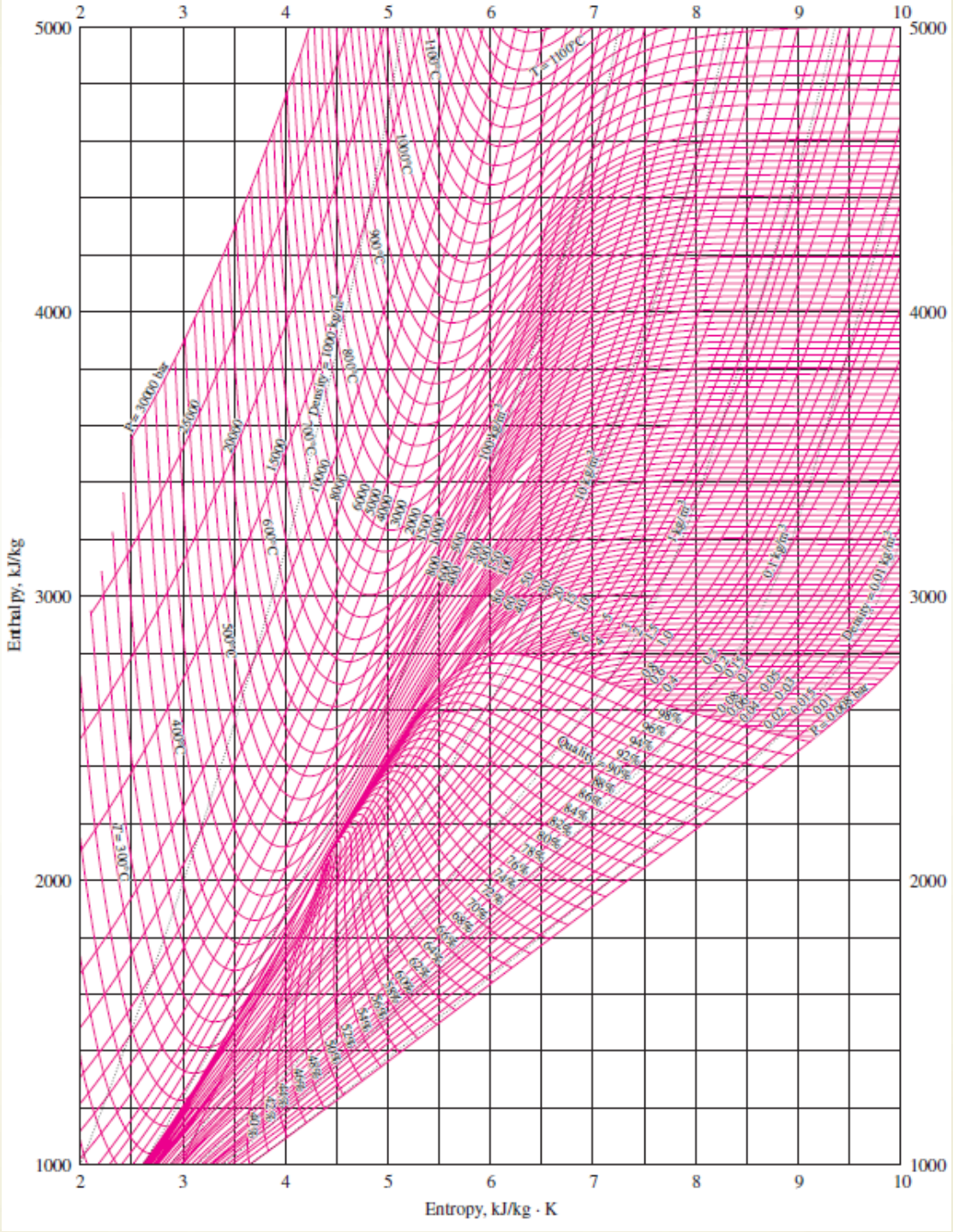
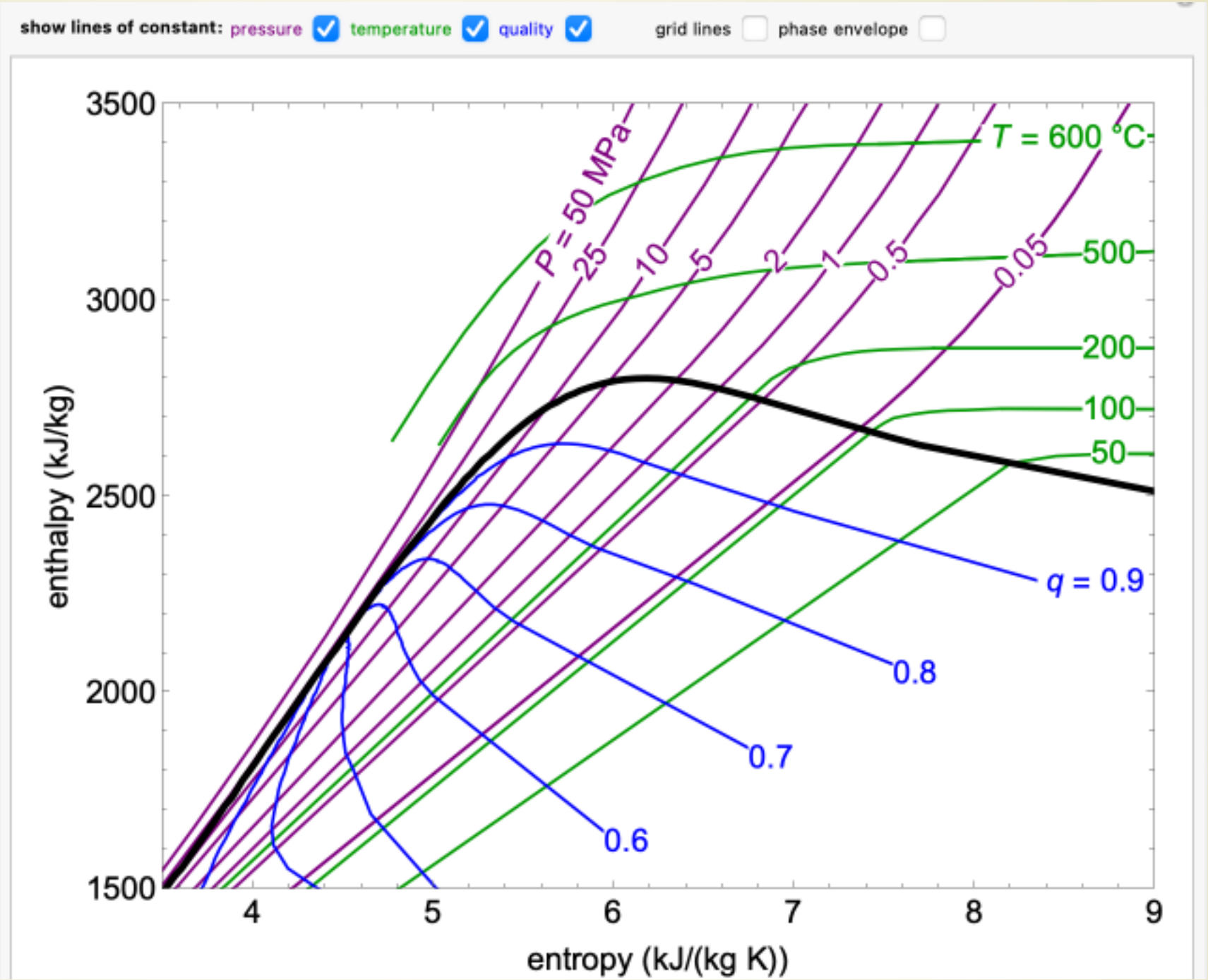


FIGURE A-10

Mollier diagram for water.

Mollier diagram for water.



EXAMPLE 7–6 The *T-S* Diagram of the Carnot Cycle

Show the Carnot cycle on a *T-S* diagram and indicate the areas that represent the heat supplied Q_H, heat rejected Q_L, and the net work output $W_{net,out}$ on this diagram.

SOLUTION The Carnot cycle is to be shown on a *T-S* diagram, and the areas that represent Q_H, Q_L, and $W_{net,out}$ are to be indicated.

Analysis Recall that the Carnot cycle is made up of two reversible isothermal (T = constant) processes and two isentropic (s = constant) processes. These four processes form a rectangle on a *T-S* diagram, as shown in Fig. 7–19.

On a *T-S* diagram, the area under the process curve represents the heat transfer for that process. Thus the area *A*12*B* represents Q_H, the area *A*43*B* represents Q_L, and the difference between these two (the area in color) represents the net work since

$$W_{net,out} = Q_H - Q_L$$

Therefore, the area enclosed by the path of a cycle (area 1234) on a *T-S* diagram represents the net work. Recall that the area enclosed by the path of a cycle also represents the net work on a *P-V* diagram.

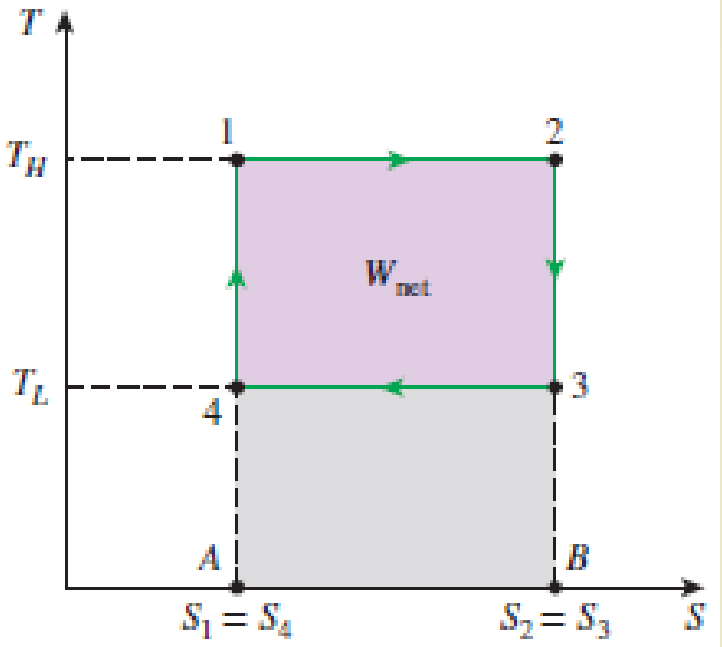


FIGURE 7–19
The *T-S* diagram of a Carnot cycle (Example 7–6).

***A12B* represents Q_H**

***A43B* represents Q_L**

1234 represents the net work

7.7 The *T ds* Relation

- The quantity $(\delta Q/T)_{\text{int rev}}$ corresponds to a differential change in entropy.
- The entropy change for a process can be evaluated by integrating $\delta Q/T$ along some imaginary internally reversible path between the actual end states.
- The differential form of the conservation of energy equation for a closed stationary system (a fixed mass) containing a simple compressible substance can be expressed for an internally reversible process as

$$\delta Q_{\text{int rev}} - \delta W_{\text{int rev,out}} = dU$$

but

$$\delta Q_{\text{int rev}} = T\,dS$$

$$\delta W_{\text{int rev,out}} = P\,dV$$

- Thus,

$$T\,dS = dU + P\,dV \qquad \text{(kJ)}$$

Or

$$T\,ds = du + P\,dv \qquad \text{(kJ/kg)}$$

- This equation is known as **the first *T ds* equation**, or **Gibbs' equation**.
- The only work interaction is the boundary work.

7.7 The *T ds* Relation (2)

- The second *T ds* equation is obtained by eliminating *du* from the last equation by using the definition of enthalpy ($h = u + Pv$):

$$\left.\begin{array}{lcl} h = u + Pv & \longrightarrow & dh = du + P\,dv + v\,dP \\ \text{(Eq. 7-23)} & \longrightarrow & T\,ds = du + P\,dv \end{array}\right\} T\,ds = dh - v\,dP$$

- These equations are extremely valuable since they relate entropy changes of a system to the changes in other properties.
- Explicit relations for differential changes in entropy:

$$ds = \frac{du}{T} + \frac{P\,dv}{T}$$

and

$$ds = \frac{dh}{T} - \frac{v\,dP}{T}$$

- The entropy change during a process can be determined by integrating either of these equations between the initial and the final states.

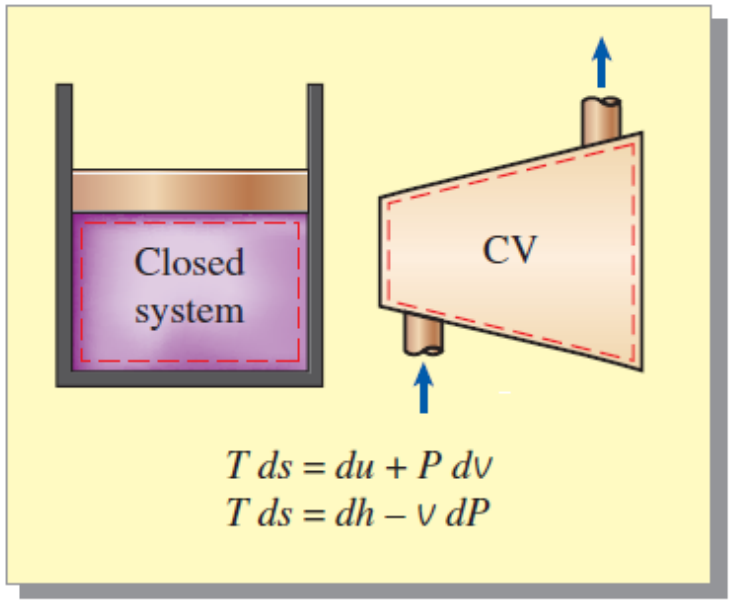


FIGURE 7–27
The *T ds* relations are valid for both reversible and irreversible processes and for both closed and open systems.

7.8 Entropy Change of Liquids and Solids

- Liquids and solids can be approximated as incompressible substances since their specific volumes remain nearly constant during a process.
- Thus $dv \cong 0$ for liquids and solids, and the entropy change equation reduces to

$$ds = \frac{du}{T} = \frac{c\,dT}{T}$$

$$ds = \frac{du}{T} + \frac{P\,dv}{T}$$

since $c_p = c_v = c$ and $du = c\,dT$ for incompressible substances.

- Then the entropy change during a process is

$$s_2 - s_1 = \int_1^2 c(T)\frac{dT}{T} \cong c_{avg} \ln\frac{T_2}{T_1} \qquad (\text{kJ/kg·K})$$

Liquids, solids

$$s_2 - s_1 = c_{avg} \ln\frac{T_2}{T_1} = 0 \longrightarrow T_2 = T_1$$

Isentropic

where c_{avg} is the average specific heat of the substance over the given temperature interval.

EXAMPLE 7–7 Effect of Density of a Liquid on Entropy

Liquid methane is commonly used in various cryogenic applications. The critical temperature of methane is 191 K (or –82°C), and thus methane must be maintained below 191 K to keep it in liquid phase. The properties of liquid methane at various temperatures and pressures are given in Table 7–1. Determine the entropy change of liquid methane as it undergoes a process from 110 K and 1 MPa to 120 K and 5 MPa (*a*) using tabulated properties and (*b*) approximating liquid methane as an incompressible substance. What is the error involved in the latter case?

SOLUTION Liquid methane undergoes a process between two specified states. The entropy change of methane is to be determined by using actual data and by assuming methane to be incompressible.

Analysis (*a*) We consider a unit mass of liquid methane (Fig. 7–28). The properties of the methane at the initial and final states are

State 1:

$$\left.\begin{aligned} P_1 &= 1\ \text{MPa} \\ T_1 &= 110\ \text{K} \end{aligned}\right\} \begin{aligned} s_1 &= 4.875\ \text{kJ/kg·K} \\ c_{p1} &= 3.471\ \text{kJ/kg·K} \end{aligned}$$

State 2:

$$\left.\begin{aligned} P_2 &= 5\ \text{MPa} \\ T_2 &= 120\ \text{K} \end{aligned}\right\} \begin{aligned} s_2 &= 5.145\ \text{kJ/kg·K} \\ c_{p2} &= 3.486\ \text{kJ/kg·K} \end{aligned}$$

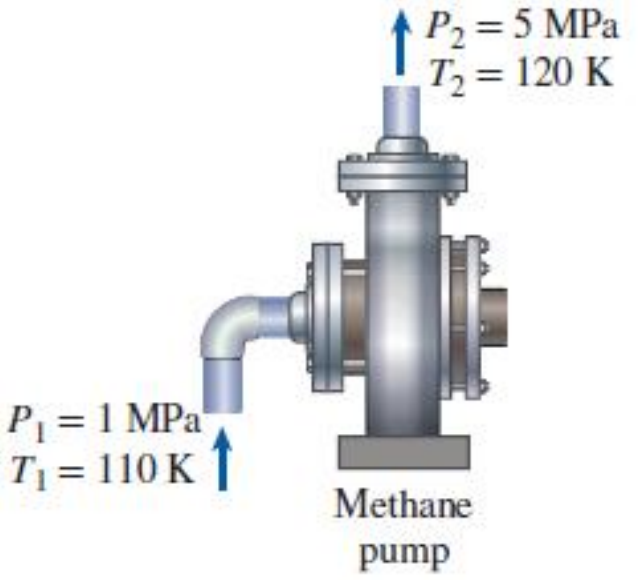


FIGURE 7–28
Schematic for Example 7–7.

TABLE 7–1
Properties of liquid methane

Temp., T, K	Pressure, P, MPa	Density, ρ, kg/m³	Enthalpy, h, kJ/kg	Entropy, s, kJ/kg·K	Specific heat, c_p, kJ/kg·K
110	0.5	425.3	208.3	4.878	3.476
	1.0	425.8	209.0	4.875	3.471
	2.0	426.6	210.5	4.867	3.460
	5.0	429.1	215.0	4.844	3.432
120	0.5	410.4	243.4	5.185	3.551
	1.0	411.0	244.1	5.180	3.543
	2.0	412.0	245.4	5.171	3.528
	5.0	415.2	249.6	5.145	3.486

Therefore,

$$\Delta s = s_2 - s_1 = 5.145 - 4.875 = \mathbf{0.270\ kJ/kg{\cdot}K}$$

(*b*) Approximating liquid methane as an incompressible substance, its entropy change is determined to be

$$\Delta s = c_{avg} \ln \frac{T_2}{T_1} = (3.4785\ \text{kJ/kg·K}) \ln \frac{120\ \text{K}}{110\ \text{K}} = \mathbf{0.303\ kJ/kg{\cdot}K}$$

since

$$c_{avg} = \frac{c_{p1} + c_{p2}}{2} = \frac{3.471 + 3.486}{2} = 3.4785\ \text{kJ/kg·K}$$

Therefore, the error involved in approximating liquid methane as an incompressible substance is

$$\text{Error} = \frac{|\Delta s_{actual} - \Delta s_{ideal}|}{\Delta s_{actual}} = \frac{|0.270 - 0.303|}{0.270} = \mathbf{0.122}\ (\text{or } 12.2\%)$$

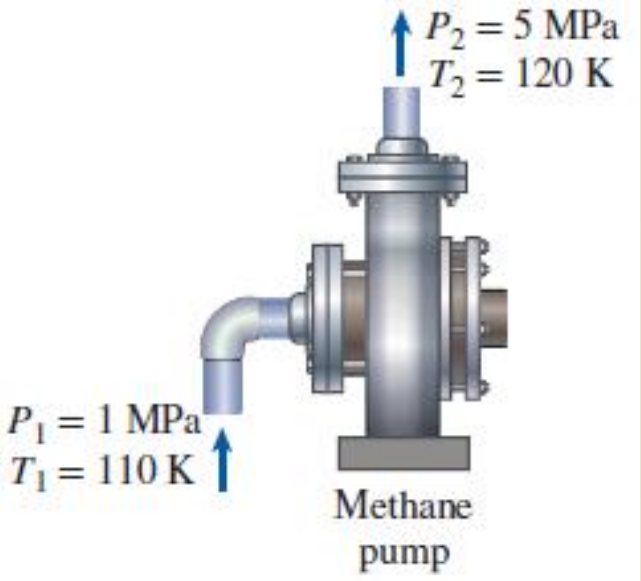


FIGURE 7–28
Schematic for Example 7–7.

Discussion This result is not surprising since the density of liquid methane changes during this process from 425.8 to 415.2 kg/m^3 (about 3 percent), which makes us question the validity of the incompressible substance assumption. Still, this assumption enables us to obtain reasonably accurate results with less effort, which proves to be very convenient in the absence of compressed liquid data.

7.9 The Entropy Change of Ideal Gases

- The entropy change for an ideal gas can be obtained by employing the property relations for ideal gases.
- Substituting $du = c_v\, dT$ and $P = RT/v$ into the entropy equation, the differential entropy change of an ideal gas becomes

$$ds = c_v \frac{dT}{T} + R\frac{dv}{v}$$

$$ds = \frac{du}{T} + \frac{P\,dv}{T}$$

- The entropy change for a process is obtained by integrating this relation:

$$s_2 - s_1 = \int_1^2 c_v(T)\frac{dT}{T} + R\ln\frac{v_2}{v_1}$$

- A second relation for the entropy change of an ideal gas is obtained by substituting $dh = c_p\, dT$ and $v = RT/P$ and integrating:

$$s_2 - s_1 = \int_1^2 c_p(T)\frac{dT}{T} - R\ln\frac{P_2}{P_1}$$

$$ds = \frac{dh}{T} - \frac{v\,dP}{T}$$

7.9 The Entropy Change of Ideal Gases (2)

Constant Specific Heats (Approximate Analysis)

- The specific heats in the equations depend on temperature and can be approximated by assuming constant specific heats for ideal gas.
- Thus, $c_v(T)$ and $c_p(T)$ are replaced with $c_{v,avg}$ and $c_{p,avg}$, respectively, and performing the integrations:

$$s_2 - s_1 = c_{v,\text{avg}} \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \qquad \text{(kJ/kg·K)}$$

and

$$s_2 - s_1 = c_{p,\text{avg}} \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \qquad \text{(kJ/kg·K)}$$

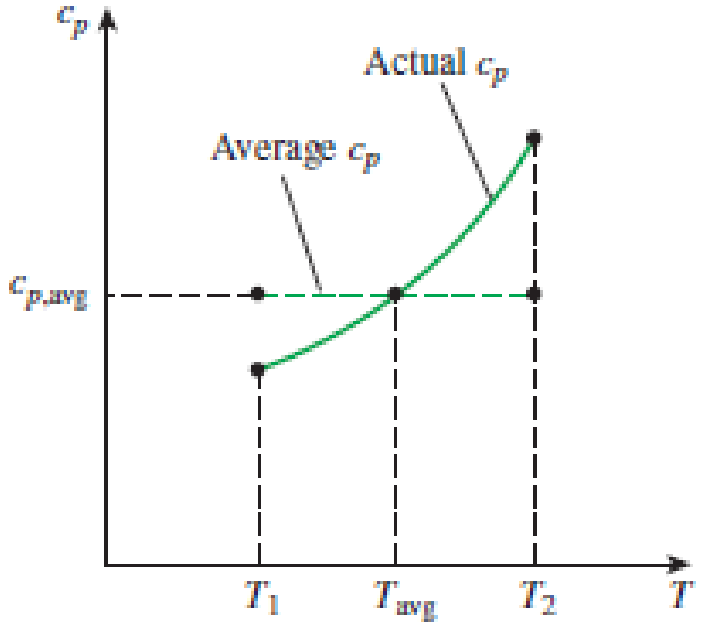


FIGURE 7–31
Under the constant-specific-heat assumption, the specific heat is assumed to be constant at some average value.

7.9 The Entropy Change of Ideal Gases (3)

Variable Specific Heats (Exact Analysis)

- When the temperature change during a process is large and the specific heats of the ideal gas vary nonlinearly within the temperature range, the assumption of constant specific heats may lead to considerable errors in entropy-change calculations.
- Instead of performing these laborious integrals each time we have a new process, we choose absolute zero as the reference temperature and define a function $s°$ as

$$s^\circ = \int_0^T c_p(T)\,\frac{dT}{T}$$

- $s°$ is a function of temperature alone, and its value is zero at absolute zero temperature.

7.9 The Entropy Change of Ideal Gases (4)

- The values of $s°$ are calculated at various temperatures, and the results are tabulated in the appendix as a function of temperature for air. Thus,

$$\int_1^2 c_p(T)\,\frac{dT}{T} = s_2^\circ - s_1^\circ$$

where $s°_2$ is the value of $s°$ at T_2 and $s°_1$ is the value at T_1. So,

On a unit–mass basis

$$s_2 - s_1 = s_2^\circ - s_1^\circ - R\ln\frac{P_2}{P_1} \qquad (\text{kJ/kg}\cdot\text{K})$$

On a unit–mole basis

$$\bar{s}_2 - \bar{s}_1 = \bar{s}_2^\circ - \bar{s}_1^\circ - R_u\ln\frac{P_2}{P_1} \qquad (\text{kJ/kmol}\cdot\text{K})$$

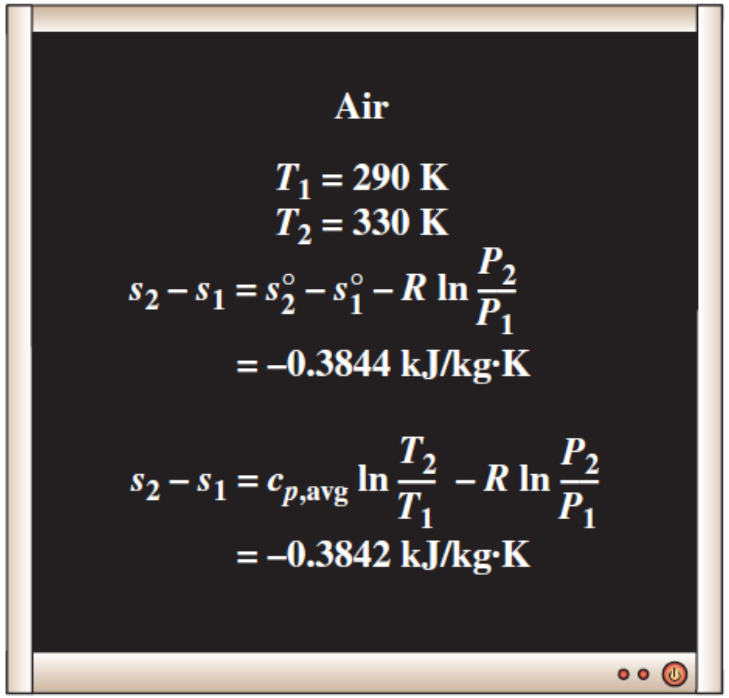


FIGURE 7–34
For small temperature differences, the exact and approximate relations for entropy changes of ideal gases give almost identical results.

T, K	$s°$, kJ/kg·K
⋮	⋮
300	1.70203
310	1.73498
320	1.76690
⋮	⋮

(Table A-17)

FIGURE 7–32
The entropy of an ideal gas depends on both T and P. The function $s°$ represents only the temperature-dependent part of entropy.

EXAMPLE 7–9 Entropy Change of an Ideal Gas

Air is compressed from an initial state of 100 kPa and 17°C to a final state of 600 kPa and 57°C. Determine the entropy change of air during this compression process by using (*a*) property values from the air table and (*b*) average specific heats.

SOLUTION Air is compressed between two specified states. The entropy change of air is to be determined by using tabulated property values and also by using average specific heats.

Assumptions Air is an ideal gas since it is at a high temperature and low pressure relative to its critical-point values. Therefore, entropy change relations developed under the ideal-gas assumption are applicable.

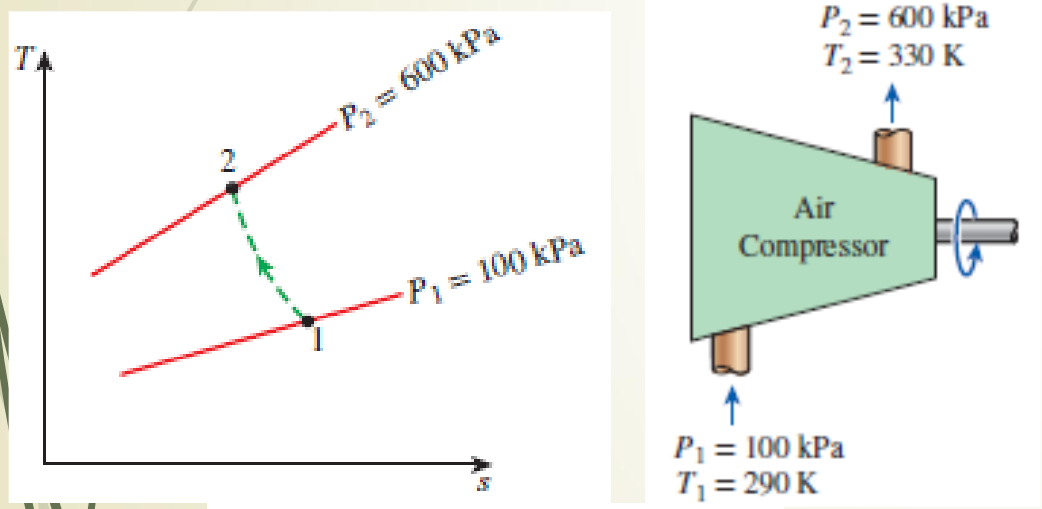


FIGURE 7–33
Schematic and *T-s* diagram for Example 7–9.

Analysis A sketch of the system and the *T-s* diagram for the process are given in Fig. 7–33. We note that both the initial and the final states of air are completely specified.

(*a*) The properties of air are given in the air table (Table A–17). Reading s° values at given temperatures and substituting, we find

$$s_2 - s_1 = s_2^\circ - s_1^\circ - R\ln\frac{P_2}{P_1}$$

$$= [(1.79783 - 1.66802)\ \text{kJ/kg·K}] - (0.287\ \text{kJ/kg·K})\ln\frac{600\ \text{kPa}}{100\ \text{kPa}}$$

$$= \mathbf{-0.3844\ kJ/kg·K}$$

(*b*) The entropy change of air during this process can also be determined approximately from Eq. 7–34 by using a c_p value at the average temperature of 37°C (Table A–2*b*) and treating it as a constant:

$$s_2 - s_1 = c_{p,\text{avg}}\ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1}$$

$$= (1.006\ \text{kJ/kg·K})\ln\frac{330\ \text{K}}{290\ \text{K}} - (0.287\ \text{kJ/kg·K})\ln\frac{600\ \text{kPa}}{100\ \text{kPa}}$$

$$= \mathbf{-0.3842\ kJ/kg·K}$$

Discussion The two results above are almost identical since the change in temperature during this process is relatively small (Fig. 7–34). When the temperature change is large, however, they may differ significantly. For those cases, Eq. 7–39 should be used instead of Eq. 7–34 since it accounts for the variation of specific heats with temperature.

7.9 The Entropy Change of Ideal Gases (5)

Isentropic Processes of Ideal Gases - Constant Specific Heats (Approximate Analysis)

- When the constant-specific-heat assumption is valid, the isentropic relations for ideal gases are obtained by setting the previous equations equal to zero

$$s_2 - s_1 = c_{v,\text{avg}} \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \longrightarrow \ln \frac{T_2}{T_1} = -\frac{R}{c_v} \ln \frac{v_2}{v_1}$$

- Rearrange:

$$\ln \frac{T_2}{T_1} = \ln \left(\frac{v_1}{v_2} \right)^{R/c_v} \longrightarrow \left(\frac{T_2}{T_1} \right)_{s = \text{const.}} = \left(\frac{v_1}{v_2} \right)^{k-1} \quad \text{(ideal gas)}$$

since $R = c_p - c_v$, $k = c_p / c_v$, and thus $R/c_v = k - 1$.

- This is the first **isentropic relation** for ideal gases under the constant-specific-heat assumption.

7.9 The Entropy Change of Ideal Gases (6)

- The second isentropic relation is obtained in a similar manner

$$s_2 - s_1 = c_{p,\text{avg}} \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \longrightarrow \left(\frac{T_2}{T_1}\right)_{s = \text{const.}} = \left(\frac{P_2}{P_1}\right)^{(k-1)/k} \quad \text{(ideal gas)}$$

- The third isentropic relation is obtained by equating the two equations

$$\left(\frac{P_2}{P_1}\right)_{s = \text{const.}} = \left(\frac{v_1}{v_2}\right)^k \quad \text{(ideal gas)}$$

- These three equations can also be expressed in a compact form as

$$Tv^{k-1} = \text{constant}$$
$$TP^{(1-k)/k} = \text{constant}$$
$$Pv^k = \text{constant}$$

7.9 The Entropy Change of Ideal Gases (7)

Isentropic Processes of Ideal Gases - Variable Specific Heats (Exact Analysis)

- the isentropic relations for ideal gases are

$$0 = s_2^\circ - s_1^\circ - R \ln \frac{P_2}{P_1} \longrightarrow s_2^\circ = s_1^\circ + R \ln \frac{P_2}{P_1}$$

- Rearrange: $\frac{P_2}{P_1} = \exp \frac{s_2^\circ - s_1^\circ}{R} \longrightarrow \frac{P_2}{P_1} = \frac{\exp(s_2^\circ / R)}{\exp(s_1^\circ / R)}$

- The quantity $C_1 \exp(s^\circ/R)$ is defined as the **relative pressure** P_r where C_1 is a constant.
- With this definition, the last relation becomes $\left(\frac{P_2}{P_1}\right)_{s=\text{const.}} = \frac{P_{r2}}{P_{r1}}$

- The relative pressure P_r is a dimensionless quantity that is a function of temperature is tabulated against temperature in appendix.

7.9 The Entropy Change of Ideal Gases (8)

- Sometimes specific volume ratios are given instead of pressure ratios.
- Utilizing the ideal gas relation,

$$\frac{P_1 v_1}{T_1} = \frac{P_2 v_2}{T_2} \rightarrow \frac{v_2}{v_1} = \frac{T_2}{T_1}\frac{P_1}{P_2} = \frac{T_2}{T_1}\frac{P_{r1}}{P_{r2}} = \frac{T_2/P_{r2}}{T_1/P_{r1}}$$

- The quantity C_2T/P_r is a function of temperature only and is defined as **relative specific volume** v_r where C_2 is a constant. Thus,

$$\left(\frac{v_2}{v_1}\right)_{s=\text{const.}} = \frac{v_{r2}}{v_{r1}}$$

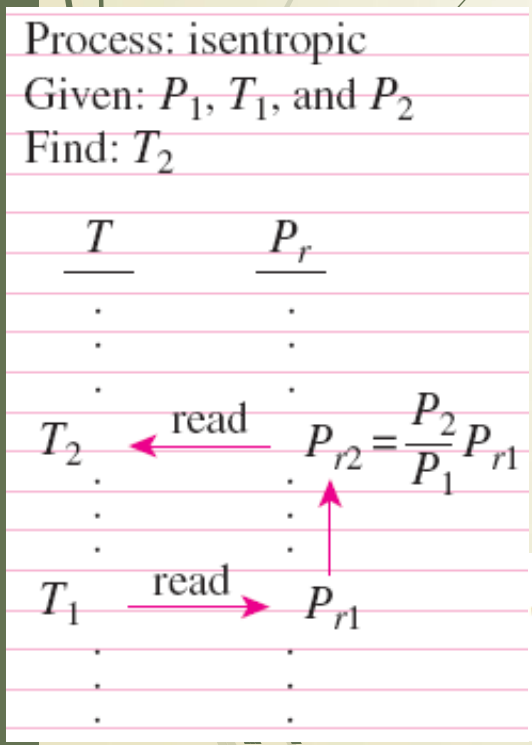


FIGURE 7–36
The use of P_r data for calculating the final temperature during an isentropic process.

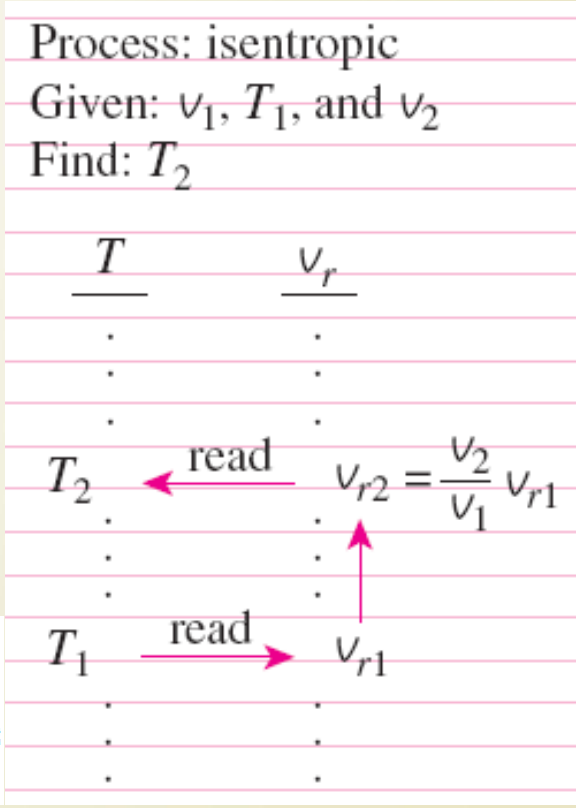


FIGURE 7–38
The use of v_r data for calculating the final temperature during an isentropic process (Example 7–10).

EXAMPLE 7–10 Isentropic Compression of Air in a Car Engine

Air is compressed in a car engine from 22°C and 95 kPa in a reversible and adiabatic manner. If the compression ratio V_1/V_2 of this engine is 8, determine the final temperature of the air.

SOLUTION Air is compressed in a car engine isentropically. For a given compression ratio, the final air temperature is to be determined.

Assumptions At specified conditions, air can be treated as an ideal gas. Therefore, the isentropic relations for ideal gases are applicable.

Analysis A sketch of the system and the *T-s* diagram for the process are given in Fig. 7–37.

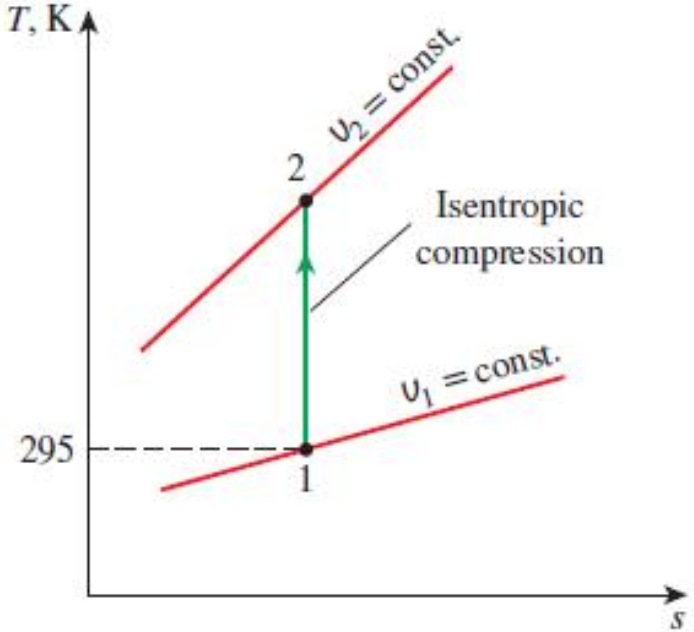
This process is easily recognized as being isentropic since it is both reversible and adiabatic. The final temperature for this isentropic process can be determined from Eq. 7–50 with the help of relative specific volume data (Table A–17), as illustrated in Fig. 7–38.

For closed systems: $$\frac{V_2}{V_1} = \frac{v_2}{v_1}$$

At $T_1 = 295$ K: $$v_{r1} = 647.9$$

From Eq. 7-50: $$v_{r2} = v_{r1}\left(\frac{v_2}{v_1}\right) = (647.9)\left(\frac{1}{8}\right) = 80.99 \quad \rightarrow \quad T_2 = \mathbf{662.7\ K}$$

Therefore, the temperature of air will increase by 367.7°C during this process.



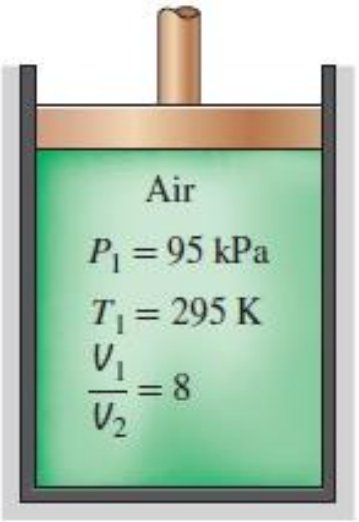


FIGURE 7–37
Schematic and *T-s* diagram for Example 7–10.

ALTERNATIVE SOLUTION The final temperature could also be determined from Eq. 7–42 by assuming constant specific heats for air:

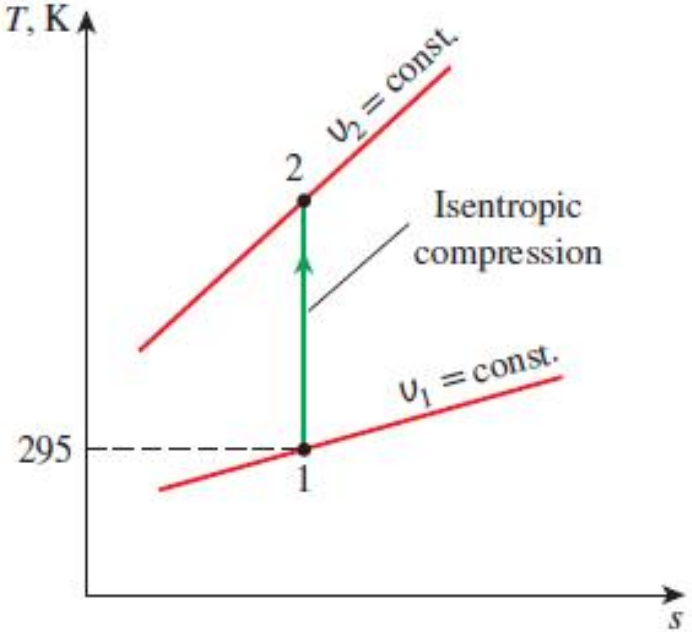
$$\left(\frac{T_2}{T_1}\right)_{s=\text{const.}} = \left(\frac{v_1}{v_2}\right)^{k-1}$$

The specific heat ratio k also varies with temperature, and we need to use the value of k corresponding to the average temperature. However, the final temperature is not given, and so we cannot determine the average temperature in advance. For such cases, calculations can be started with a k value at the initial or the anticipated average temperature. This value could be refined later, if necessary, and the calculations can be repeated. We know that the temperature of the air will rise considerably during this adiabatic compression process, so we *guess* the average temperature to be about 450 K. The k value at this anticipated average temperature is determined from Table A–2*b* to be 1.391. Then, the final temperature of air becomes

$$T_2 = (295\text{ K})(8)^{1.391-1} = 665.2\text{ K}$$

This gives an average temperature value of 480.1 K, which is sufficiently close to the assumed value of 450 K. Therefore, it is not necessary to repeat the calculations by using the k value at this average temperature.

The result obtained by assuming constant specific heats for this case is in error by about 0.4 percent, which is rather small. This is not surprising since the temperature change of air is relatively small (only a few hundred degrees) and the specific heats of air vary almost linearly with temperature in this temperature range.



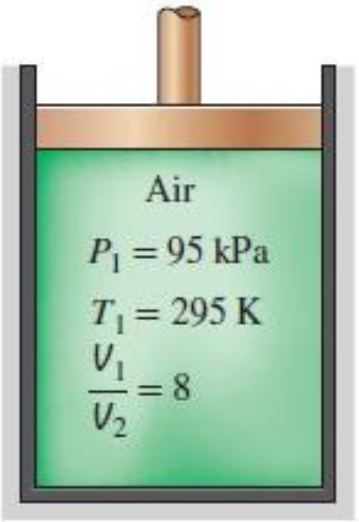


FIGURE 7–37
Schematic and *T-s* diagram for Example 7–10.

Exercise 1

7–41 A rigid tank contains 5 kg of saturated vapor steam at 100°C. The steam is cooled to the ambient temperature of 25°C.

(*a*) Sketch the process with respect to the saturation lines on a *T*-*v* diagram.

(*b*) Determine the entropy change of the steam, in kJ/K.

(*c*) For the steam and its surroundings, determine the total entropy change associated with this process, in kJ/K.

Exercise 2

7–56 A piston–cylinder device contains 5 kg of steam at 100°C with a quality of 50 percent. This steam undergoes two processes as follows:

1-2 Heat is transferred to the steam in a reversible manner while the temperature is held constant until the steam exists as a saturated vapor.

2-3 The steam expands in an adiabatic, reversible process until the pressure is 15 kPa.

(*a*) Sketch these processes with respect to the saturation lines on a single *T-s* diagram.

(*b*) Determine the heat transferred to the steam in process 1-2, in kJ.

(*c*) Determine the work done by the steam in process 2-3, in kJ.

Exercise 3

7–60 A 50-kg copper block initially at 140°C is dropped into an insulated tank that contains 90 L of water at 10°C. Determine the final equilibrium temperature and the total entropy change for this process.

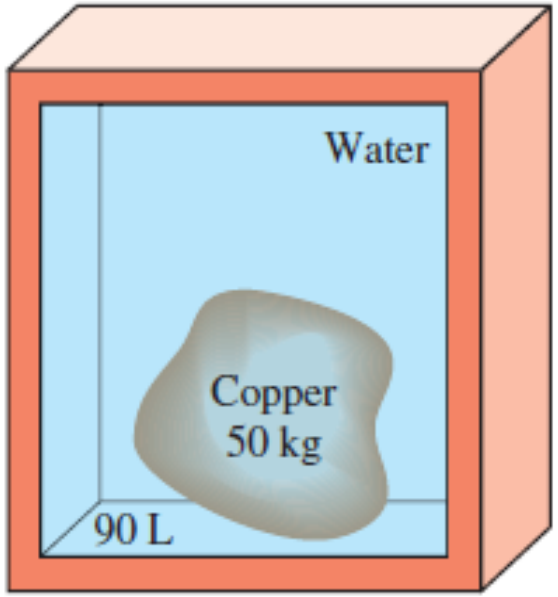


FIGURE P7–60

7.12 Isentropic Efficiencies of Steady-Flow Devices

- To examine the degree of degradation of energy in steady flow devices, first we need to define an ideal process as reference, that is the isentropic process.
- The more closely the actual process approximates the idealized isentropic process, the better the device performs.
- This idealized model cycle enabled us to:
  - determine the theoretical limits of performance for cyclic devices under specified conditions and
  - examine how the performance of actual devices suffered as a result of irreversibilities.
- Isentropic efficiency – a parameter which is a measure of the deviation of actual process from the corresponding idealized ones
- Isentropic efficiencies are defined differently for different devices since each device is set up to perform different tasks.

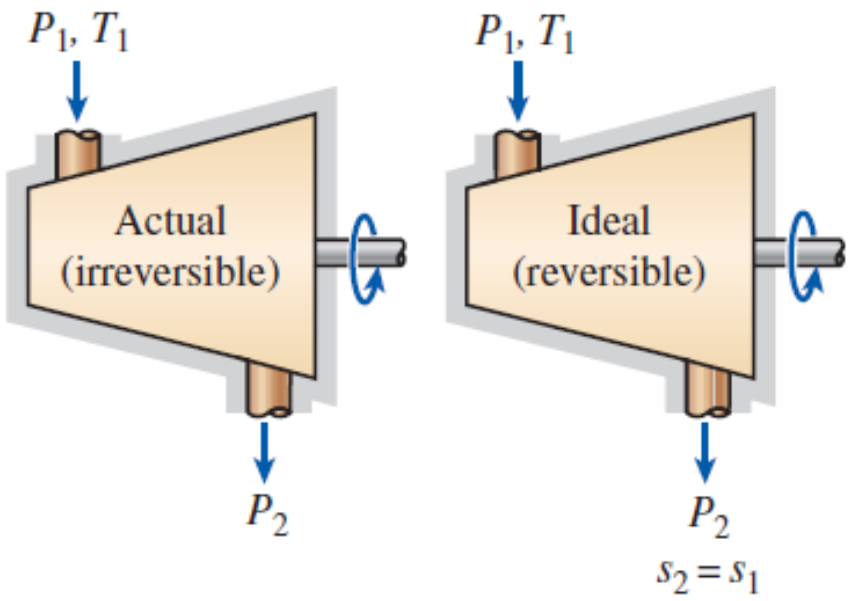


FIGURE 7–47
The isentropic process involves no irreversibilities and serves as the ideal process for adiabatic devices.

7.12 Isentropic Efficiencies of Steady-Flow Devices (2)

Isentropic Efficiency of Turbines

- For a turbine under steady operation, the inlet state of the working fluid and the exhaust pressure are fixed.
  - Therefore, the ideal process for an adiabatic turbine is an isentropic process between the inlet state and the exhaust pressure.
- The isentropic efficiency of a turbine is defined as

$$\eta_T = \frac{\text{Actual turbine work}}{\text{Isentropic turbine work}} = \frac{w_a}{w_s}$$

- When the changes in kinetic and potential energies associated with a fluid stream flowing through a turbine are small relative to the change in enthalpy and can be neglected, the work output of an adiabatic turbine simply becomes **the change in enthalpy**,

$$\eta_T \cong \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$$

Well-designed, large turbines $\eta_T > 90\%$
Small turbines, $\eta_T < 70\%$.

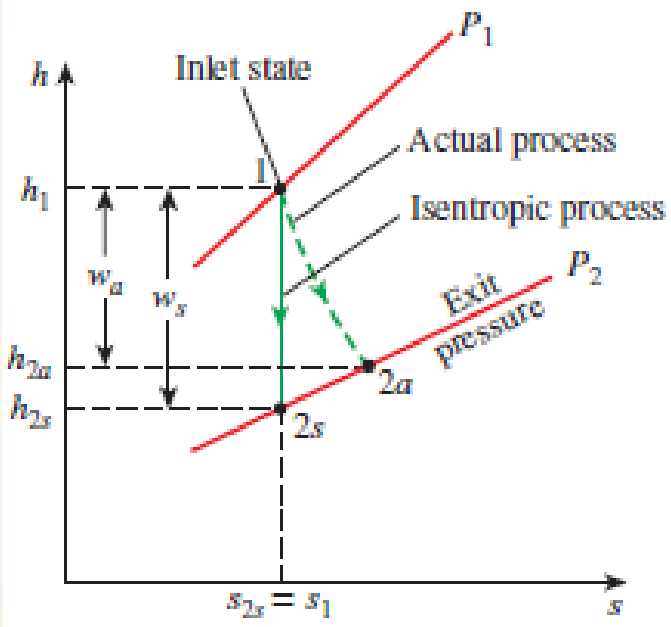


FIGURE 7–48
The *h-s* diagram for the actual and isentropic processes of an adiabatic turbine.

EXAMPLE 7–14 Isentropic Efficiency of a Steam Turbine

Steam enters an adiabatic turbine steadily at 3 MPa and 400°C and leaves at 50 kPa and 100°C. If the power output of the turbine is 2 MW, determine (*a*) the isentropic efficiency of the turbine and (*b*) the mass flow rate of the steam flowing through the turbine.

SOLUTION Steam flows steadily in a turbine between inlet and exit states. For a specified power output, the isentropic efficiency and the mass flow rate are to be determined.

Assumptions **1** Steady operating conditions exist. **2** The changes in kinetic and potential energies are negligible.

Analysis A sketch of the system and the *T-s* diagram of the process are given in Fig. 7–49.

(*a*) The enthalpies at various states are

State 1: $\left.\begin{array}{l} P_1 = 3 \text{ MPa} \\ T_1 = 400°\text{C} \end{array}\right\} \begin{array}{l} h_1 = 3231.7 \text{ kJ/kg} \\ s_1 = 6.9235 \text{ kJ/kg·K} \end{array}$ (Table A–6)

State 2a: $\left.\begin{array}{l} P_{2a} = 50 \text{ kPa} \\ T_{2a} = 100°\text{C} \end{array}\right\} h_{2a} = 2682.4 \text{ kJ/kg}$ (Table A–6)

The exit enthalpy of the steam for the isentropic process h_{2s} is determined from the requirement that the entropy of the steam remain constant ($s_{2s} = s_1$):

State 2s: $\begin{array}{l} P_{2s} = 50 \text{ kPa} \\ (s_{2s} = s_1) \end{array} \longrightarrow \begin{array}{l} s_f = 1.0912 \text{ kJ/kg·K} \\ s_g = 7.5931 \text{ kJ/kg·K} \end{array}$ (Table A–5)

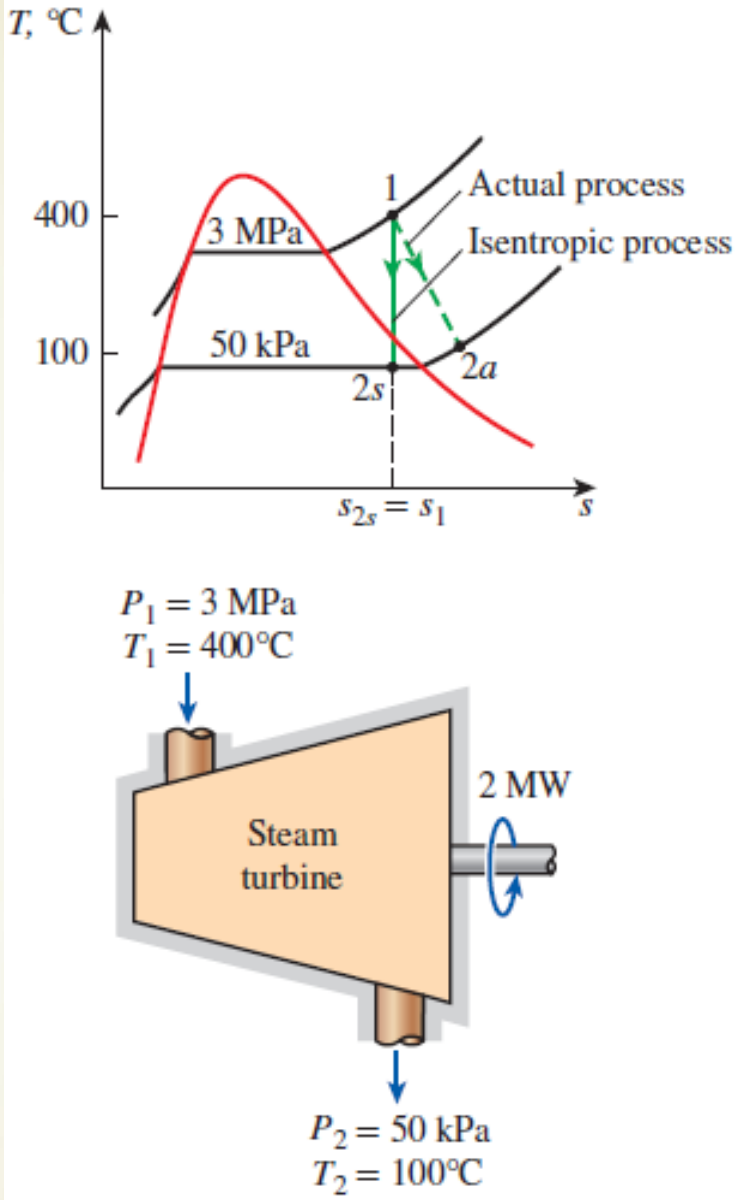


FIGURE 7–49
Schematic and *T-s* diagram for Example 7–14.

Obviously, at the end of the isentropic process steam exists as a saturated mixture since $s_f < s_{2s} < s_g$. Thus, we need to find the quality at state 2*s* first:

$$x_{2s} = \frac{s_{2s} - s_f}{s_{fg}} = \frac{6.9235 - 1.0912}{6.5019} = 0.897$$

and

$$h_{2s} = h_f + x_{2s}h_{fg} = 340.54 + 0.897(2304.7) = 2407.9 \text{ kJ/kg}$$

By substituting these enthalpy values into Eq. 7–61, the isentropic efficiency of this turbine is determined to be

$$\eta_T \cong \frac{h_1 - h_{2a}}{h_1 - h_{2s}} = \frac{3231.7 - 2682.4}{3231.7 - 2407.9} = \mathbf{0.667} \text{ (or 66.7\%)}$$

(*b*) The mass flow rate of steam through this turbine is determined from the energy balance for steady-flow systems:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}h_1 = \dot{W}_{a,out} + \dot{m}h_{2a}$$

$$\dot{W}_{a,out} = \dot{m}(h_1 - h_{2a})$$

$$2 \text{ MW}\left(\frac{1000 \text{ kJ/s}}{1 \text{ MW}}\right) = \dot{m}(3231.7 - 2682.4) \text{ kJ/kg}$$

$$\dot{m} = \mathbf{3.64 \text{ kg/s}}$$

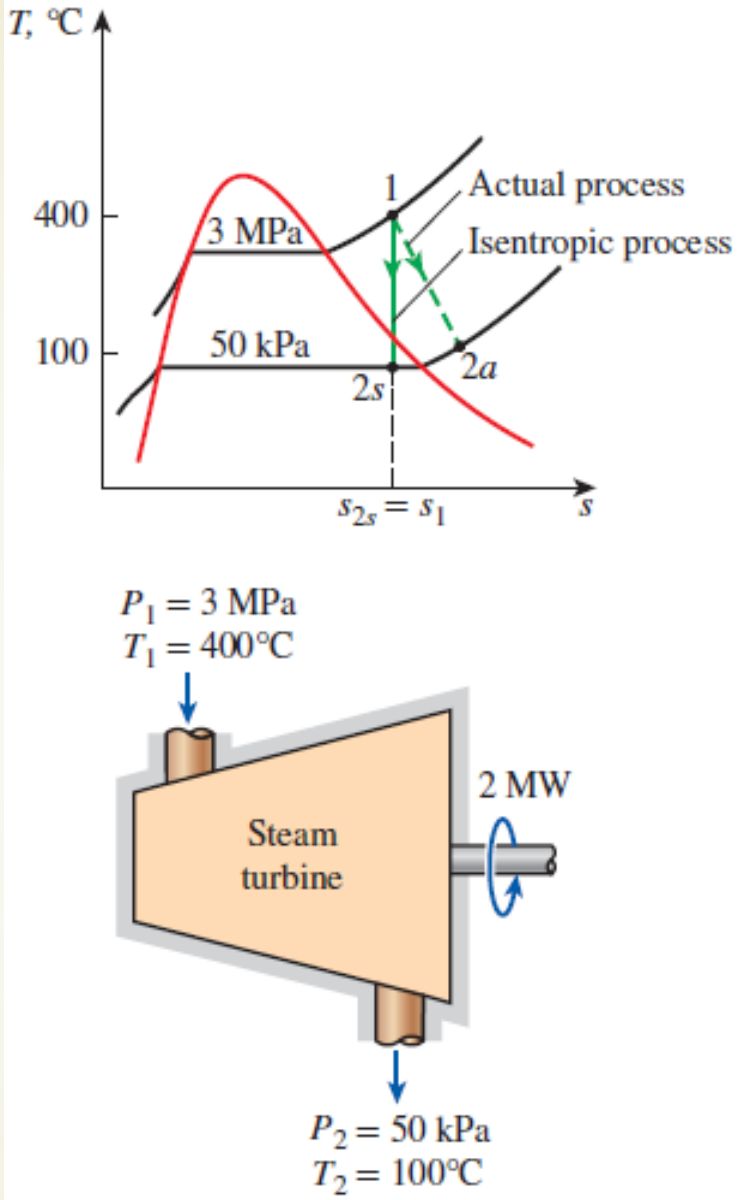


FIGURE 7–49
Schematic and *T*-*s* diagram for Example 7–14.

7.12 Isentropic Efficiencies of Steady-Flow Devices (3)

Isentropic Efficiencies of Compressors and Pumps

- The isentropic efficiency of a compressor is defined as:

the ratio of the work input required to raise the pressure of a gas to a specified value in an isentropic manner to the actual work input

$$\eta_C = \frac{\text{Isentropic compressor work}}{\text{Actual compressor work}} = \frac{w_s}{w_a}$$

- When the changes in kinetic and potential energies of the gas being compressed are negligible,

$$\eta_C \cong \frac{h_{2s} - h_1}{h_{2a} - h_1}$$

- Well-designed compressors have isentropic efficiencies that range from 80 to 90 percent.

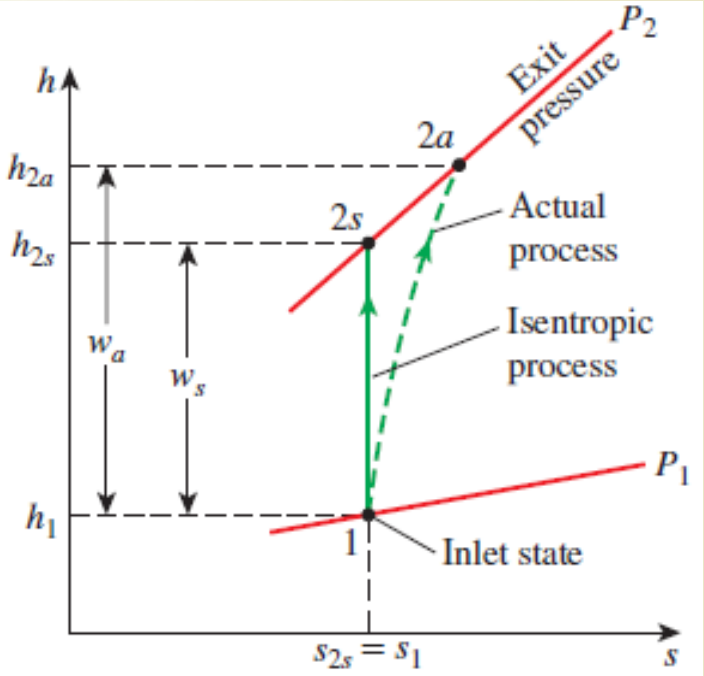


FIGURE 7–50
The *h-s* diagram of the actual and isentropic processes of an adiabatic compressor.

7.12 Isentropic Efficiencies of Steady-Flow Devices (4)

- When the changes in potential and kinetic energies of a liquid are negligible, the isentropic efficiency of a pump is

$$\eta_P = \frac{w_s}{w_a} = \frac{v(P_2 - P_1)}{h_{2a} - h_1}$$

- A realistic model process for compressors that are intentionally cooled during the compression process is the reversible isothermal process.
- Then we can define an isothermal efficiency

$$\eta_C = \frac{w_t}{w_a}$$

where w_t and w_a are the required work inputs to the compressor for the reversible isothermal and actual cases, respectively.

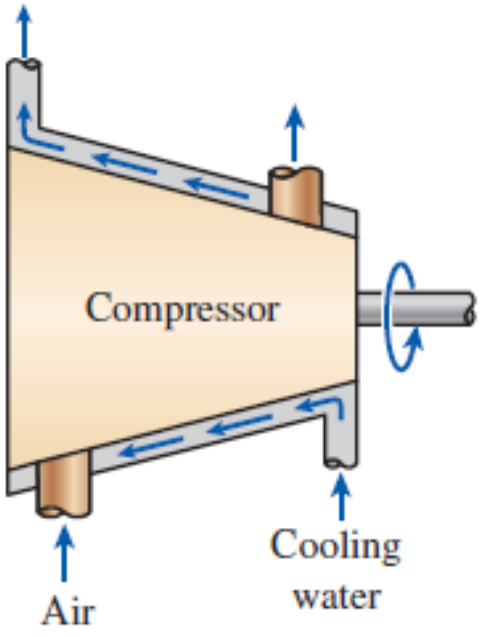


FIGURE 7–51
Compressors are sometimes intentionally cooled to minimize the work input.

EXAMPLE 7–15 Effect of Efficiency on Compressor Power Input

Air is compressed by an adiabatic compressor from 100 kPa and 12°C to a pressure of 800 kPa at a steady rate of 0.2 kg/s. If the isentropic efficiency of the compressor is 80 percent, determine (*a*) the exit temperature of air and (*b*) the required power input to the compressor.

SOLUTION Air is compressed to a specified pressure at a specified rate. For a given isentropic efficiency, the exit temperature and the power input are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas. **3** The changes in kinetic and potential energies are negligible.

Analysis A sketch of the system and the *T-s* diagram of the process are given in Fig. 7–52.

(*a*) We know only one property (pressure) at the exit state, and we need to know one more to fix the state and thus determine the exit temperature. The property that can be determined with minimal effort in this case is h_{2a} since the isentropic efficiency of the compressor is given. At the compressor inlet,

$$T_1 = 285\text{ K} \rightarrow h_1 = 285.14\text{ kJ/kg} \qquad \text{(Table A–17)}$$
$$P_{r1} = 1.1584$$

The enthalpy of the air at the end of the isentropic compression process is determined by using one of the isentropic relations of ideal gases,

$$P_{r2} = P_{r1}\left(\frac{P_2}{P_1}\right) = 1.1584\left(\frac{800\text{ kPa}}{100\text{ kPa}}\right) = 9.2672$$

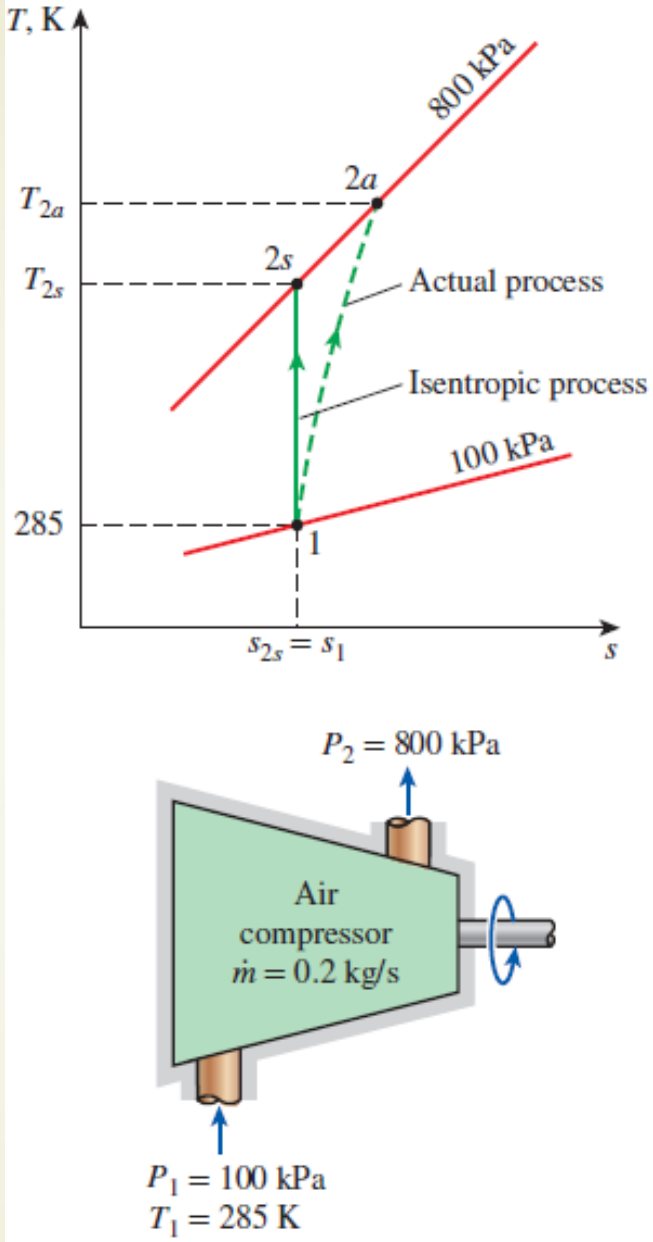


FIGURE 7–52
Schematic and *T-s* diagram for Example 7–15.

and

$$P_{r2} = 9.2672 \quad \rightarrow \quad h_{2s} = 517.05 \text{ kJ/kg}$$

Substituting the known quantities into the isentropic efficiency relation, we have

$$\eta_C \cong \frac{h_{2s} - h_1}{h_{2a} - h_1} \quad \rightarrow \quad 0.80 = \frac{(517.05 - 285.14) \text{ kJ/kg}}{(h_{2a} - 285.14) \text{ kJ/kg}}$$

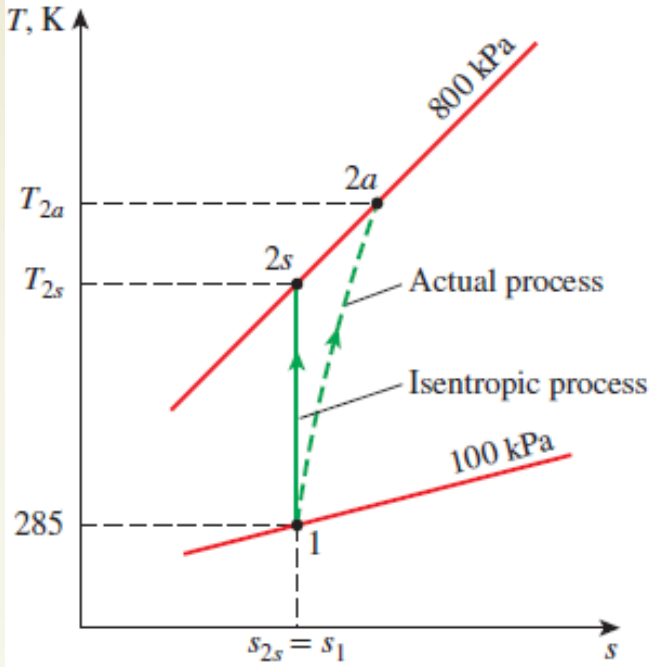
Thus,

$$h_{2a} = 575.03 \text{ kJ/kg} \quad \rightarrow \quad T_{2a} = \mathbf{569.5\ K}$$

(*b*) The required power input to the compressor is determined from the energy balance for steady-flow devices,

$$\begin{aligned} \dot{E}_{in} &= \dot{E}_{out} \\ \dot{m}h_1 + \dot{W}_{a,in} &= \dot{m}h_{2a} \\ \dot{W}_{a,in} &= \dot{m}(h_{2a} - h_1) \\ &= (0.2 \text{ kg/s})[(575.03 - 285.14) \text{ kJ/kg}] \\ &= \mathbf{58.0\ kW} \end{aligned}$$

Discussion Notice that in determining the power input to the compressor, we used h_{2a} instead of h_{2s} since h_{2a} is the actual enthalpy of the air as it exits the compressor. The quantity h_{2s} is a hypothetical enthalpy value that the air would have if the process were isentropic.



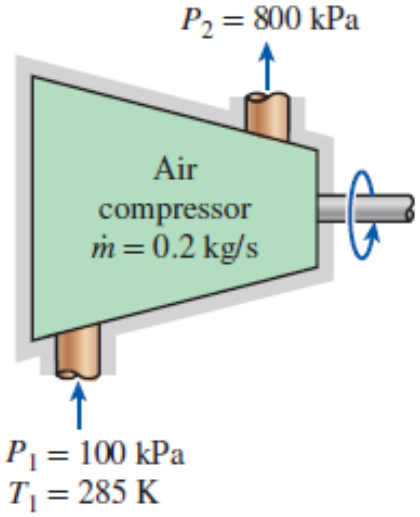


FIGURE 7–52
Schematic and *T-s* diagram for Example 7–15.

7.12 Isentropic Efficiencies of Steady-Flow Devices (5)

Isentropic Efficiency of Nozzles

- Nozzles are essentially adiabatic devices and are used to accelerate a fluid.
- Isentropic efficiency of a nozzle is defined as

the ratio of the actual kinetic energy of the fluid at the nozzle exit to the kinetic energy value at the exit of an isentropic nozzle for the same inlet state and exit pressure

$$\eta_N = \frac{\text{Actual KE at nozzle exit}}{\text{Isentropic KE at nozzle exit}} = \frac{V_{2a}^2}{V_{2s}^2}$$

- Isentropic efficiencies of nozzles are typically above 90%
  - nozzle efficiencies above 95% are not uncommon.

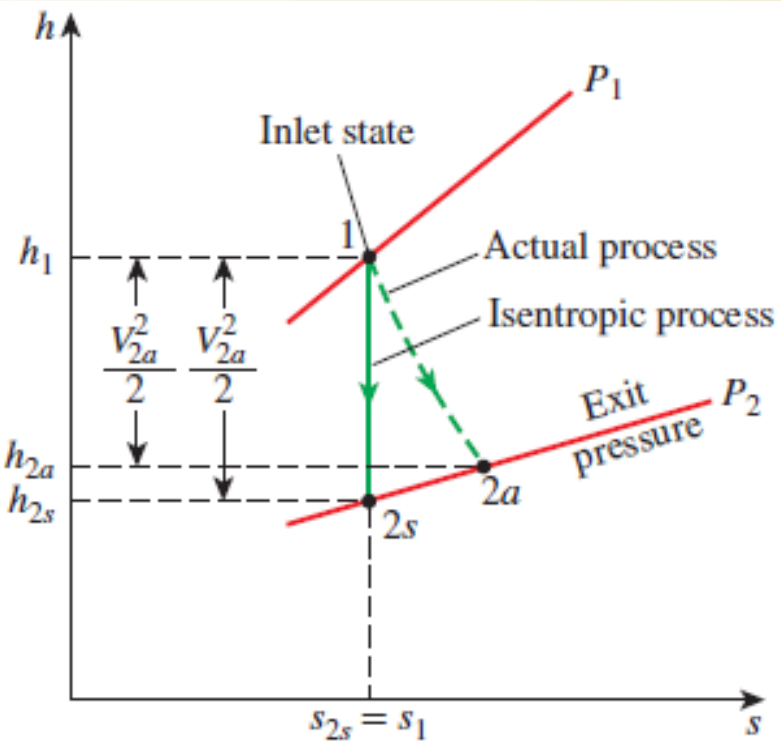


FIGURE 7–53
The *h-s* diagram of the actual and isentropic processes of an adiabatic nozzle.

7.12 Isentropic Efficiencies of Steady-Flow Devices (4)

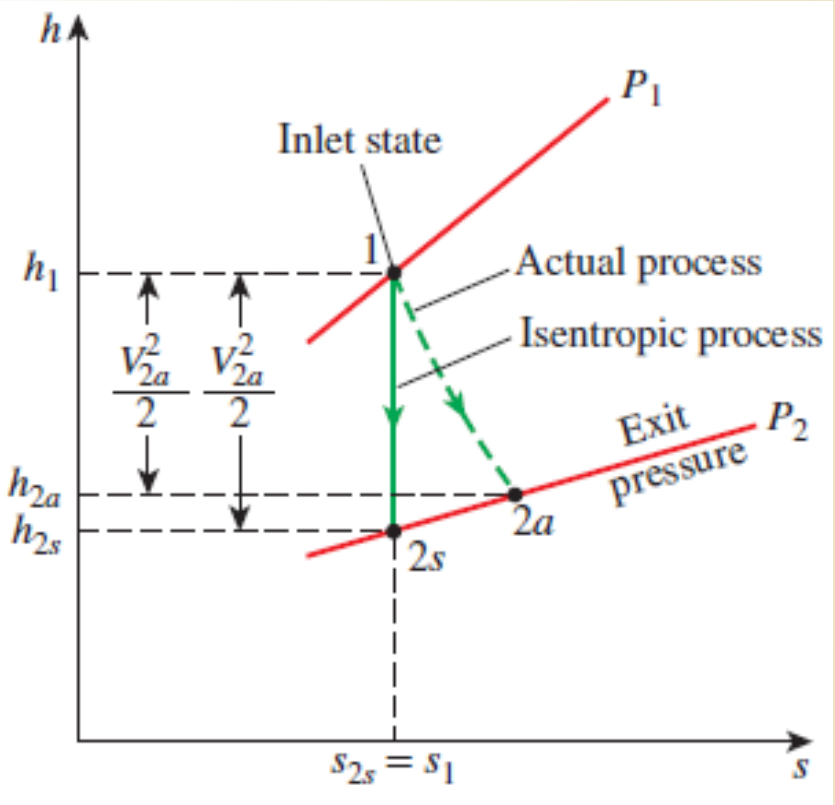
- Nozzles involve no work interactions, little or no change in its potential energy and the inlet velocity of the fluid is small relative to the exit velocity
- Thus, the energy balance for this steady-flow device reduces to

$$h_1 = h_{2a} + \frac{V_{2a}^2}{2}$$

- Then the isentropic efficiency of the nozzle can be expressed in terms of enthalpies as

$$\eta_N \cong \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$$

where h_{2a} and h_{2s} are the enthalpy values at the nozzle exit for the actual and isentropic processes, respectively.



EXAMPLE 7–16 Effect of Efficiency on Nozzle Exit Velocity

Air at 200 kPa and 950 K enters an adiabatic nozzle at low velocity and is discharged at a pressure of 110 kPa. If the isentropic efficiency of the nozzle is 92 percent, determine (*a*) the maximum possible exit velocity, (*b*) the exit temperature, and (*c*) the actual exit velocity of the air. Assume constant specific heats for air.

SOLUTION The acceleration of air in a nozzle is considered. For specified exit pressure and isentropic efficiency, the maximum and actual exit velocities and the exit temperature are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas. **3** The inlet kinetic energy is negligible.

Analysis A sketch of the system and the *T-s* diagram of the process are given in Fig. 7–54.

The temperature of air will drop during this acceleration process because some of its internal energy is converted to kinetic energy. This problem can be solved accurately by using property data from the air table. But we will assume constant specific heats (thus sacrifice some accuracy) to demonstrate their use. Let us guess the average temperature of the air to be about 850 K. Then, the average values of c_p and k at this anticipated average temperature are determined from Table A–2*b* to be c_p = 1.11 kJ/kg·K and k = 1.349.

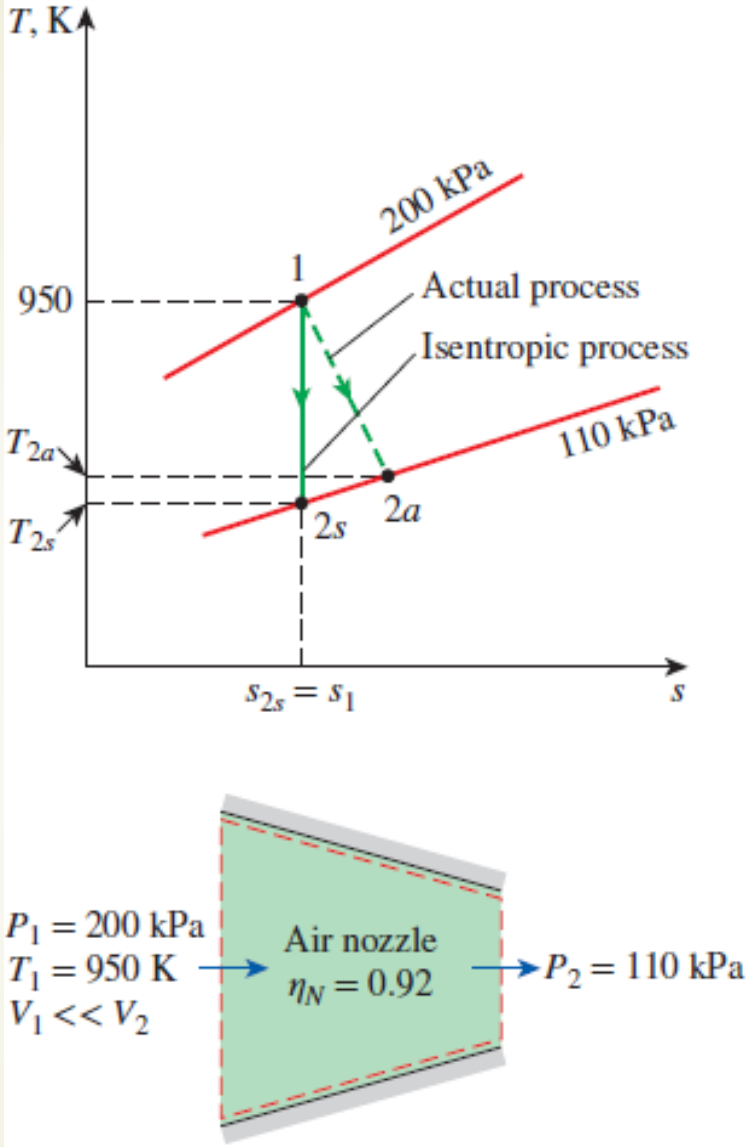


FIGURE 7–54
Schematic and *T-s* diagram for Example 7–16.

(*a*) The exit velocity of the air will be a maximum when the process in the nozzle involves no irreversibilities. The exit velocity in this case is determined from the steady-flow energy equation. However, first we need to determine the exit temperature. For the isentropic process of an ideal gas we have:

$$\frac{T_{2s}}{T_1} = \left(\frac{P_{2s}}{P_1}\right)^{(k-1)/k}$$

or

$$T_{2s} = T_1\left(\frac{P_{2s}}{P_1}\right)^{(k-1)/k} = (950\text{ K})\left(\frac{110\text{ kPa}}{200\text{ kPa}}\right)^{0.349/1.349} = 814\text{ K}$$

This gives an average temperature of 882 K, which is somewhat higher than the assumed average temperature (850 K). This result could be refined by reevaluating the k value at 882 K and repeating the calculations, but it is not warranted since the two average temperatures are sufficiently close (doing so would change the temperature by only 0.6 K, which is not significant).

Now we can determine the isentropic exit velocity of the air from the energy balance for this isentropic steady-flow process:

$$e_{\text{in}} = e_{\text{out}}$$

$$h_1 + \frac{V_1^2}{2} = h_{2s} + \frac{V_{2s}^2}{2}$$

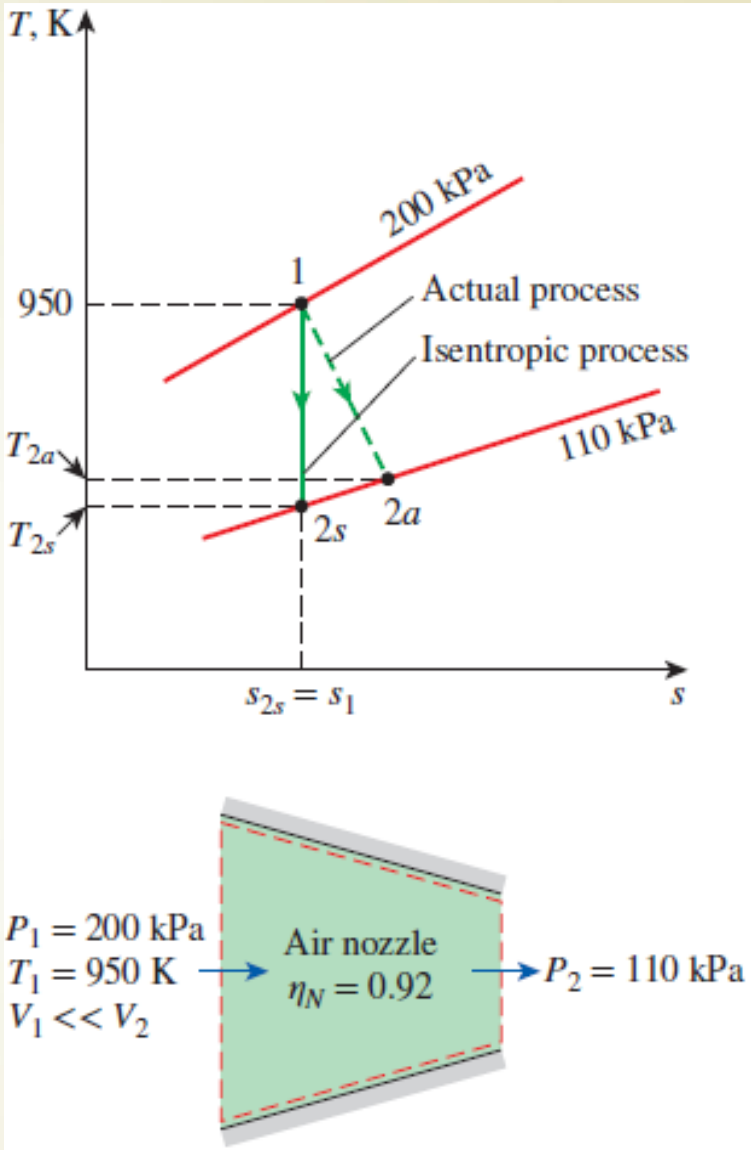


FIGURE 7–54
Schematic and *T-s* diagram for Example 7–16.

or

$$V_{2s} = \sqrt{2(h_1 - h_{2s})} = \sqrt{2c_{p,\text{avg}}(T_1 - T_{2s})}$$
$$= \sqrt{2(1.11\ \text{kJ/kg·K})[(950 - 814)\text{K}]\left(\frac{1000\ \text{m}^2/\text{s}^2}{1\ \text{kJ/kg}}\right)}$$
$$= \mathbf{549\ m/s}$$

(*b*) The actual exit temperature of the air is higher than the isentropic exit temperature evaluated above and is determined from

$$\eta_N \cong \frac{h_1 - h_{2a}}{h_1 - h_{2s}} = \frac{c_{p,\text{avg}}(T_1 - T_{2a})}{c_{p,\text{avg}}(T_1 - T_{2s})}$$

or

$$0.92 = \frac{950 - T_{2a}}{950 - 814} \rightarrow T_{2a} = \mathbf{825\ K}$$

That is, the temperature is 11 K higher at the exit of the actual nozzle as a result of irreversibilities such as friction. It represents a loss since this rise in temperature comes at the expense of kinetic energy (Fig. 7–55).

(*c*) The actual exit velocity of air can be determined from the definition of isentropic efficiency of a nozzle,

$$\eta_N = \frac{V_{2a}^2}{V_{2s}^2} \rightarrow V_{2a} = \sqrt{\eta_N V_{2s}^2} = \sqrt{0.92(549\ \text{m/s})^2} = \mathbf{527\ m/s}$$

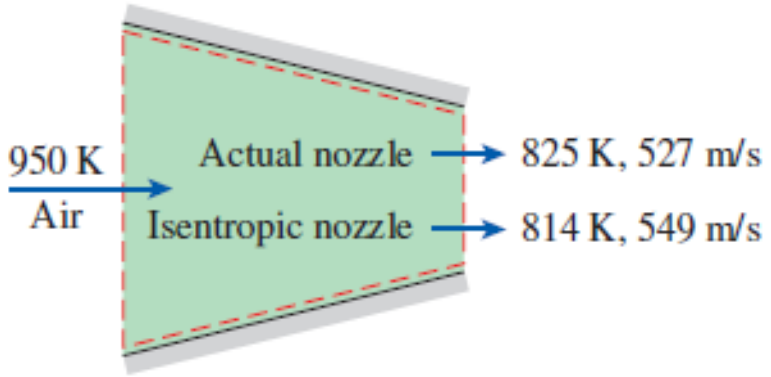


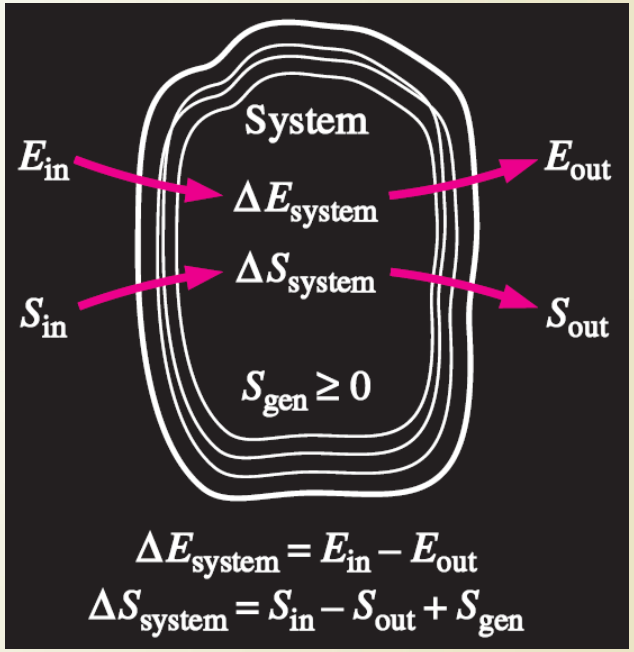
FIGURE 7–55
A substance leaves actual nozzles at a higher temperature (thus a lower velocity) as a result of friction.

7.13 Entropy Balance

- Entropy is a measure of molecular disorder or randomness of a system, and the second law of thermodynamics states that entropy can be created but it cannot be destroyed.
- the increase of entropy principle for any system is expressed as

$$\begin{pmatrix}\text{Total}\\ \text{entropy}\\ \text{entering}\end{pmatrix} - \begin{pmatrix}\text{Total}\\ \text{entropy}\\ \text{leaving}\end{pmatrix} + \begin{pmatrix}\text{Total}\\ \text{entropy}\\ \text{generated}\end{pmatrix} = \begin{pmatrix}\text{Change in the}\\ \text{total entropy}\\ \text{of the system}\end{pmatrix}$$

or $$S_{in} - S_{out} + S_{gen} = \Delta S_{system}$$



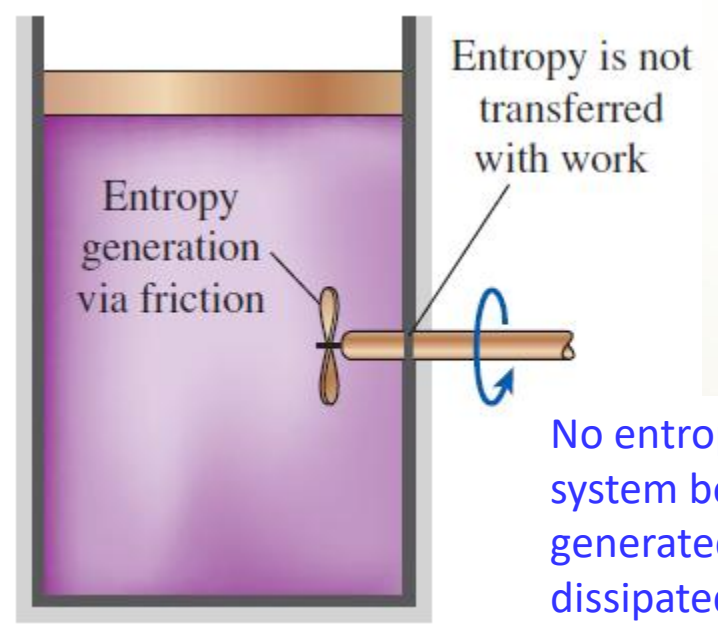
- This relation is often referred to as the entropy balance and is applicable to any system undergoing any process, which can be stated as

the entropy change of a system during a process is equal to the net entropy transfer through the system boundary plus the entropy generated within the system.

7.13 Entropy Balance (2)

Mechanisms of Entropy Transfer, S_{in} and S_{out}

- Entropy can be transferred to or from a system by two mechanisms:
  - Heat transfer and
  - Mass flow
- The only form of entropy interaction in a fixed mass or closed system is heat transfer, and thus the entropy transfer for an **adiabatic closed system is zero**.



No entropy accompanies work as it crosses the system boundary. But entropy may be generated within the system as work is dissipated into a less useful form of energy.

7.13 Entropy Balance (3)

- Heat transfer mechanism:
  - Entropy flow or entropy transfer by **heat transfer** is expressed as

$$S_{heat} = \frac{Q}{T} \quad (T = \text{constant})$$

  - the direction of entropy transfer is the same as the direction of heat transfer
- Mass flow mechanism:
  - When a mass *m* enters or leaves a system, entropy in the amount of *ms*, accompanies it, that is,

$$S_{mass} = ms$$

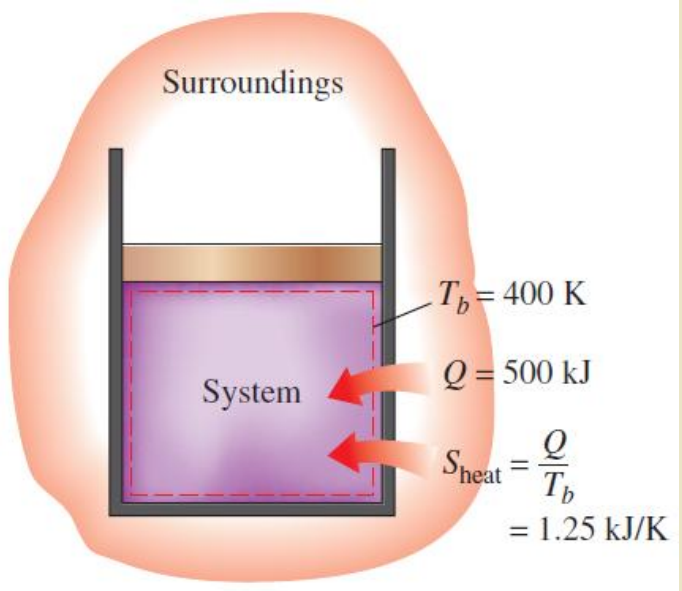


FIGURE 7–57
Heat transfer is always accompanied by entropy transfer in the amount of Q/T, where T is the boundary temperature.

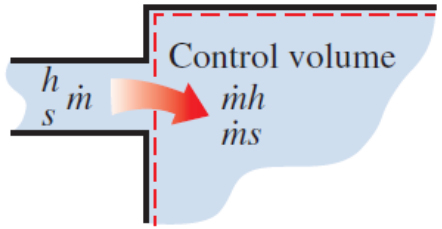


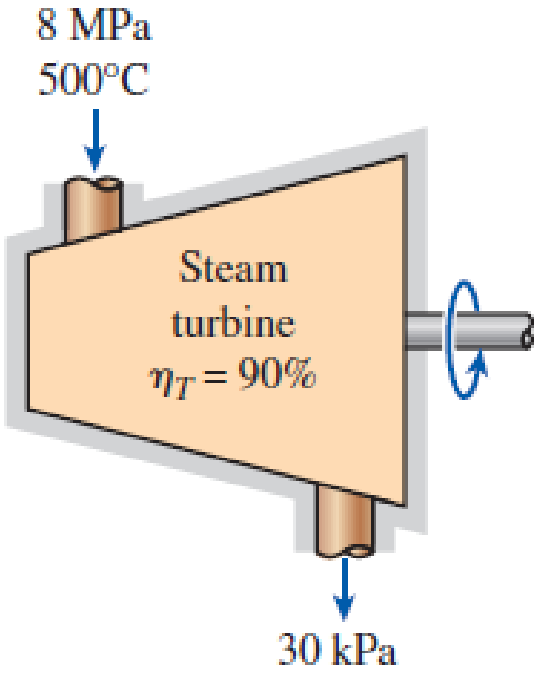
FIGURE 7–59
Mass contains entropy as well as energy, and thus mass flow into or out of system is always accompanied by energy and entropy transfer.

Exercise 4

7–117 Steam enters an adiabatic turbine at 5 MPa, 650°C, and 80 m/s and leaves at 50 kPa, 150°C, and 140 m/s. If the power output of the turbine is 8 MW, determine (*a*) the mass flow rate of the steam flowing through the turbine and (*b*) the isentropic efficiency of the turbine. *Answers:* (*a*) 8.03 kg/s, (*b*) 82.8 percent

Exercise 5

7–120 Steam enters an adiabatic turbine at 8 MPa and 500°C with a mass flow rate of 3 kg/s and leaves at 30 kPa. The isentropic efficiency of the turbine is 0.90. Neglecting the kinetic energy change of the steam, determine (*a*) the temperature at the turbine exit and (*b*) the power output of the turbine. *Answers:* (*a*) 69.1°C, (*b*) 3054 kW



Exercise 6

7–126 Air is compressed by an adiabatic compressor from 95 kPa and 27°C to 600 kPa and 277°C. Assuming variable specific heats and neglecting the changes in kinetic and potential energies, determine (*a*) the isentropic efficiency of the compressor and (*b*) the exit temperature of air if the process were reversible. *Answers:* (*a*) 81.9 percent, (*b*) 506 K

Exercise 7

7–134 Oxygen enters an insulated 12-cm-diameter pipe with a velocity of 70 m/s. At the pipe entrance, the oxygen is at 240 kPa and 20°C; and, at the exit, it is at 200 kPa and 18°C. Calculate the rate at which entropy is generated in the pipe.

Thank You